

CMMaTH: A Chinese Multi-modal Math Skill Evaluation Benchmark for Foundation Models

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Abstract

With the rapid advancements in multimodal large language models, evaluating their multimodal mathematical capabilities continues to receive wide attention. Although datasets such as MathVista have been introduced for evaluating mathematical capabilities in multimodal scenarios, there remains a lack of evaluation tools and datasets tailored for fine-grained assessment in Chinese K12 education. To systematically evaluate the ability of multimodal large models to solve Chinese multimodal mathematical problems, we propose a Chinese Multi-modal Math Skill Evaluation Benchmark (CMMaTH), containing 23,856 multimodal K12 math related questions, making it the largest Chinese multimodal mathematical problem benchmark to date. CMMaTH includes questions ranging from elementary to high school levels, offering greater diversity in problem types, solution goals, visual elements, detailed knowledge points, and standard solution annotations. To facilitate stable, fast, and cost-free model evaluation, we have developed an open-source tool called GradeGPT, which is integrated with the CMMaTH dataset. Our data and code are available at <https://github.com/zzli2022/CMMaTH>.

1 Introduction

Large language models (LLMs) excel in various language tasks, while Large Multimodal Models (LMMs) effectively handle visual-language problems. They drive advancements in natural language processing and computer vision, offering powerful solutions for complex tasks, demonstrating potential as versatile solvers for multimodal problems.

The systematic evaluation of LLM/LMMs' performance across various mathematical reasoning scenarios has been an extensive research subject. GSM8K and MATH (Cobbe et al., 2021;

Hendrycks et al., 2021b) assessed the ability for multi-step mathematical reasoning by constructing a high-quality set of elementary school math word problems or various competition mathematics problems. By collecting a diverse set of mathematical problems containing both textual and visual components, Lu et al. (2023), Wang et al. (2024), Zhang et al. (2024c) systematically evaluated the ability of LMMs to perceive visual elements and solve corresponding multimodal problems. Shi et al. (2023a) constructed a multilingual mathematical reasoning dataset, MGSM, for evaluating the LLM reasoning ability in multilingual environments. However, in non-English multimodal contexts, especially in Chinese scenarios, there is still a lack of sufficiently detailed and diverse benchmarks for assessing mathematical abilities. To assess the capability of large language models in non-English contexts, Huang et al. (2023) and Zhang et al. (2024b) constructed multidisciplinary Chinese question answering datasets C-Eval and CMMMU to evaluate the knowledge and reasoning abilities of LMMs. However, C-Eval lacks evaluation in multimodal contexts, while CMMMU's dataset has relatively low diversity, consisting of only 540 questions.

Meanwhile, existing datasets contain numerous problem scenarios that are not typically examined in real K12 education. There are inconsistencies between the assessment results and the actual abilities of students in K12 mathematics. Liu et al. (2024b) has introduced a benchmark for mathematics assessment, which includes questions with detailed annotations of textbook knowledge points. The goal is to address these discrepancies in real K12 educational settings. However, this benchmark is limited to text-only scenarios and does not include multimodal questions and knowledge points.

Existing math benchmarks for answer evaluation can be categorized into two types: *rule-based* (Cobbe et al. (2021), Hendrycks et al. (2021b), He

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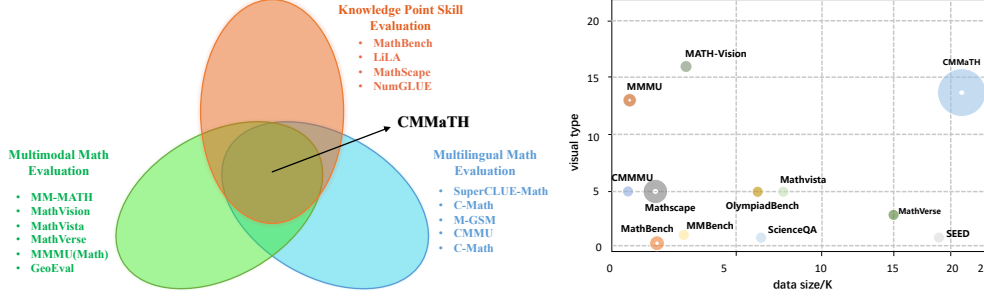


Figure 1: The CMMaTH dataset focuses on the multi-modal mathematical ability assessment of Chinese scenes and has extremely strong diversity and large data scale. CMMaTH combines fine-grained multi-modal mathematical knowledge point annotation. The size of the hollow circle corresponds to the designed number of knowledge points in the dataset.

Dataset	Language Domain	Knowledge Annotation	Knowledge Domain	Knowledge Point Number	Size	Modality	Source	Answer
VQAv2(Goyal et al., 2017)	En	×	General	—	> 1M	V+T	Annotated	Open/MC/TF
SEED(Li et al., 2023a)	En	×	General	—	19K	V+T	Annotated	MC
MMBench(Liu et al., 2023b)	En	×	General	—	3K	V+T	Repurposed	MC
MM-Vet(Yu et al., 2023)	En	×	General	—	0.2K	V+T	Annotated	Open
ScienceQA(Lu et al., 2022)	En	×	—	Science	6K	V+T	Textbooks	MC
MMMU(Yue et al., 2023)	En	×	General	—	11.5K	V(30 Types)+OC	Textbooks	Open/MC
CM3U(Zhang et al., 2024b)	ZH	×	General	—	< 1K(Math Part)	V(5 Types)+OC	Internet	Open/MC
MathVista(Lu et al., 2023)	ZH/En	×	Math	—	1K/6K	V(5 Types)+OC	Synthesized	Open/MC/TF
OlympiadBench(He et al., 2024)	ZH/En	×	Math/Physics	—	6.5K(Math Part)	V(5 Types)	Internet	Open
MathVerse(Zhang et al., 2024c)	ZH/En	×	Math	—	2.6K/15K	V(3 Types)	Synthesized	MC
MATH-Vision(Wang et al., 2024)	En	×	Math	—	3K	V(16 Types)+IC	Synthesized	Open/MC
MM-MATH(Sun et al., 2024)	ZH/En	×	Math	—	5.9K	V(5 Types)	Internet	Open
MathBench(Liu et al., 2024b)	ZH/En	✓	Math	60	3.7K	Text-only	Internet	Open
MathScape(Zhou et al., 2024a)	ZH	✓	K12 Math	107	1.3K	V	Photo-based Paper	Open
CMMaTH	ZH	✓	K12 Math	784	23K	V(14 Types), OC, IC	Internet&Annotated	Open/MC

Table 1: Comparison with other multimodal benchmarks. V: visual input, OC: optical characters caption, T: Question Text, IC: Image Caption, Open: open questions, MC: multiple choice questions, TF: true or false questions.

et al. (2024) and *API-based* methods Lu et al. (2023), Zhang et al. (2024c), Hendrycks et al. (2021a). *API-based* methods leverage the zero-shot in-context learning capabilities of LLM. However, these methods are associated with high costs and significant time consumption, often leading to unstable and inconsistent evaluation outcomes. *Rule-based* methods, aim to derive the correct option from model responses through the design of various extraction systems. Nonetheless, these methods face challenges in handling highly diverse contents of benchmarks. Also, it is difficult to maintain handcrafted rules for dynamically updated benchmarks. Current multimodal math benchmark evaluations often resort to multiple-choice or true/false question formats, using rules or API-based LLM to extract options for assessing answers.

Based on the above considerations, we propose a new multimodal mathematical benchmark CM-MaTH. Compared to previous benchmarks, our benchmark demonstrates greater diversity and increased depth of reasoning in the Chinese multimodal math context. It also includes finer-grained knowledge annotation to grasp different levels and types of K12 math knowledge. We provide an open-source lightweight answer comparator called GradeGPT, designed to compare the consistency

between outputs from different LLM/LMMs and standard answers, thus avoiding expensive evaluation costs. Leveraging the CMMaTH dataset and GradeGPT tool, we evaluate mainstream open-source and commercial LMMs in Table 3, reporting comprehensive evaluation results along with various and extensive case analyses, and knowledge skill analyses. In summary, our paper makes the following contributions:

- We develop a high-quality multimodal mathematics benchmark specifically tailored for the Chinese language context, featuring detailed knowledge point annotations, extensive quantity, and diversity. It serves as a reference for evaluating the multimodal mathematical reasoning capabilities of foundational models within Chinese language contexts. We also provide an English version of this dataset.
- Compared to previous multimodal mathematical benchmarks, our dataset exhibits a great depth of reasoning and diversity. Our benchmark simulates more realistic educational Q&A scenarios, encompassing a wider variety of question types and answer formats. Additionally, we provide each question with detailed multimodal knowledge points to eval-

uate the mastery level of current large models. The CMMaTH dataset is dynamically maintained and will be periodically updated.

- We build an evaluation assistant named GradeGPT on the CMMaTH dataset, which enables for comparing the proximity of model responses to standard answers and assessing the correctness of results. GradeGPT features lightweight open-source characteristics, avoiding the instability and high costs associated with commercial models.
- We conduct a systematic evaluation of existing mainstream LLM/LMMs, quantitatively and qualitatively comparing with existing models.

2 Related Work

2.1 Assessment of mathematical abilities

To evaluate the performance of LLM/LMMs in mathematical reasoning and examine hallucinations during the reasoning process, numerous benchmarks (Liu et al., 2023b; Sun et al., 2024; Yue et al., 2023; Yu et al., 2023; Huang et al., 2024b; Bi et al., 2024c) have been proposed for evaluating the mathematical reasoning capabilities of large models. GSM8K (Cobbe et al., 2021) is the first and most widely used mathematical dataset used for large model math evaluation, consisting of 1k math word problem test samples and corresponding answers. The MATH (Hendrycks et al., 2021b) dataset, in comparison to GSM8K, presents a greater challenge in terms of reasoning difficulty. This dataset demands a more profound understanding and intuition in various mathematical domains such as Algebra, Number Theory, and Geometry. MathVista (Lu et al., 2023) is the first dataset used to evaluate the multimodal mathematical capabilities of large models, but it has relatively simple reasoning depth. MATH-VISION (Wang et al., 2024) has richer visual elements and deeper reasoning difficulty. MathVerse (Zhang et al., 2024e) constructs several subsets of datasets to assess whether existing multimodal large models can truly understand mathematical abstract forms. MathBench (Liu et al., 2024b) attempts to assess the level of mastery of specific mathematical skills in existing large models, but this work focuses solely on the pure text domain and annotates a relatively coarse of knowledge points. MM-MATH (Sun et al., 2024) is similar to our work but primarily focuses on English scenarios. The CMMaTH Benchmark, in

comparison to existing works on the evaluation of mathematical proficiency, places a greater emphasis on the analysis of mathematical abilities within the context of the Chinese language. The data distribution of the CMMaTH dataset more closely aligns with the actual distribution found in K12 educational settings, and it provides detailed annotations of mathematical knowledge points to facilitate the assessment of models’ mastery of knowledge and skills.

2.2 Large Model Evaluation Tool

Due to their strong generalization capabilities and extensive world knowledge, large language models have achieved outstanding results in tasks such as machine translation (Zhu et al., 2023), question answering (Kamalloo et al., 2023), dialogue (Duan et al., 2023), and so on by generating text. Evaluating the comprehensive abilities of large models, such as clarity, adherence to instructions, comprehensiveness, formality, and mathematical reasoning ability, has received widespread attention (Ke et al., 2023; Mei et al., 2024c,b; Zhou et al., 2024b). Currently, many works opt to use powerful commercial model APIs, such as GPT-4, to assist in evaluating the comprehensive abilities of large models. For instance, in the field of geometric problem solving (Ning et al., 2023; Li et al., 2023c; Zhang et al., 2024d) and multimodal reasoning, MathVista (Lu et al., 2023) and GeoEval (Zhang et al., 2024c) use GPT-4’s API to extract correct answers for evaluation. These methods face several challenges: they are costly and time-consuming, and they struggle to keep up with rapid model iterations. Besides, these methods face challenges in terms of consistency and reproducibility (Wang et al., 2023a; Ke et al., 2023).

Recent methods have proposed using metrics such as BERT score (Zhang et al., 2020) or MAUVE (Pillutla et al., 2021) for evaluation. However, the numerical indicators produced by these methods are difficult to interpret when it comes to the erroneous responses generated by LLM. PandaLM and CritiqueLLM (Wang et al., 2023c; Ke et al., 2023) are similar to our work. They proposed a fine-tuning method based on open-source LLMs, distilling the evaluation capabilities of GPT-3.5 into a series of smaller open-source models. However, they are focused on the automated evaluation of more general text generation tasks, while we are targeting the automated evaluation of responses from large models for multimodal mathe-

Statistic	Number
Total questions	23856
- Choice-mode questions	18191
- Free-form questions	5665
- Questions in the testmini set	1371
Choice-mode questions	18191(76.2%)
- Single-choice questions	13706(57.4%)
- Multiple-choice & Composite questions	4485(18.8%)
Knowledge Point Number	784
Visual Subjects	13
Maximum question length	593
Minimum question length	6
Average question length	75.1
Grade Distribution Elementary(1-6)	800
Junior(7-9)	5082
Senior(10-12)	17972

Table 2: Key statistics of CMMaTH. The unit of question length is words. For more information can refer to Appendix E on the definitions of "Question Difficulty Levels," "Visual Subjects," and "Knowledge Point Number" and "Composite questions"

matics problems.

Unlike PandaLM (Wang et al., 2023c) trying to evaluate the relative conciseness, clarity, our evaluation model, GradeGPT, is a dataset-oriented answer comparator that can provide specific reasons based on the standard answer and a model’s response. We distilled the answer comparison capability of GPT-4 using the Cross-Lingual Judge-of-Chain method and enhanced GradeGPT’s answer discrimination ability.

3 CMMaTH Dataset

3.1 Overview of CMMaTH

We selected diverse multimodal mathematical problems from a vast pool of K12 educational questions, comprising 23,856 items across 14 visual themes and encompassing 784 types of knowledge points. More detailed statistical data can be found in Table 2 and Appendix E.

3.2 Collection Guidelines

We collected a large number of multimodal mathematics questions from open-source websites, which host a vast collection of K12 math problems. The quality and distribution of the data were guided by the following criteria during collection.

- **Diverse Mathematical Visual Elements.** We have collected solutions to multimodal mathematical problems that rely on understanding image content, especially those containing a

large amount of Chinese visual content such as text and symbols. Table 7 shows some visual elements subject of CMMaTH.

- **High relevance to the K12 math knowledge and skill.** The collector, being knowledgeable in the field, must ensure that each multimodal question targets a specific K12 math concept during the collection process. The dataset primarily includes K12-level math questions, enabling the evaluation of large-scale multimodal models’ potential in mathematics education.
- **High-quality images and answers.** During the collection phase, we instruct collectors to disregard multimodal math questions with erroneous symbols or low-quality images (blurry images). Collectors are required to ensure that the collected questions are generally solvable.

3.3 Data Collections

Collection from Diverse Multimodal Math Sources CMMaTH’s is mainly based on “Jiaoyan Yun”¹ and “Zujuan”². These two websites have collected a large number of real K12 education questions with test paper source annotations, grade levels and coarse-grained knowledge point annotation. We purchased the copyright for Jiaoyan Yun’s data and worked with the official website to ensure that it can be open sourced. After preliminary data collection, we compiled about 200,000 preliminary data for each grade. For more data cleaning details, refer to Appendix E.

Data Filtering We excluded all questions without images in the question stems, including those non-Chinese language questions, and those solvable without visual content. Due to OCR processing may result in inaccuracies, To ensure the quality of both images and text-based questions, we removed any images with a width or height less than 100 pixels and employed the GPT-4 API to evaluate data quality. For more details on data cleaning, refer to Appendix E.3.

Data Labeling We have adopted the current commercial knowledge graph for mathematics education, Jiaoyan Cloud, which has been validated by a large number of users and teachers. We first crawled the knowledge graph involved in "Jiaoyan Cloud", which contains 5531 knowledge points.

¹<https://www.jiaoyanyun.com/>

²<https://zujuan.xkw.com/>

数学统计图推理 Statics Map Reason ☆ 折线图 line chart ☆ 样本估计总体 Estimate the population by the sample ☆ 频率分布直方图 Frequency square distribution histogram ☆ 维恩图 Venn diagram ☆ 样本的均值与方差 Sample mean and variance ☆ 随机变量及其分布 Random variables and distribution ☆ 古典概型与几何概型 Classical and geometrical concepts ☆ 茎叶图 Stem and leaf plot ☆ 随机抽样 random sampling ☆ 众数、中位数、平均数 mode, median, mean ☆ 总体密度曲线 overall density curve	抽象图示意图推理 Abatract Sketch Map Reason ☆ 图形染色 Graphic Coloring Problem ☆ 程序框图 Block Diagram Understanding ☆ 树型图 Tree Diagram ☆ 三角形数及衍生数列 Triangular numbers and derivative sequences ☆ 周期数列 Periodic Sequence ☆ 图形排列与组合 Arrangement And Combination ☆ 加法原理与乘法原理 Principle of addition And Multiplication	函数推理 Function Reason ☆ 二次函数性质 Quadratic function properties ☆ 反比例函数的性质 Inverse proportional function properties ☆ 三角函数性质 Trigonometric properties ☆ 圆锥曲线性质 Conic properties ☆ 直线方程 Equation of a straight line
	几何推理 Geometry Reason ☆ 角的定义 Angle ☆ 三视图 Three View ☆ 棱台 Properties of prism ☆ 投影 Projection ☆ 等腰三角形 Isosceles Triangle ☆ 菱形 diamond ☆ 勾股定理 Pythagorean theorem ☆ 射影定理 Photography Theorem ☆ 三角形的内切圆性质 Triangle Inscribed Circles ☆ 角平分线的定义与性质 Angle Bisectors ☆ 垂直平分线的定义与性质 Vertical Bisector ☆ 圆内接四边形定理 Circle Inscribed Quadrilateral theorem ☆ 相似与全等 Similarity And Congruence ☆ 圆周角定理 Circle Angle Theorem	其他 Other ☆ 尺规作图 Rule and compass drawing ☆ 杨辉三角的运用 Application of Yang Hui Triangle

Figure 2: Part of the knowledge points involved in the CMMaTH dataset.

Then, we filtered out the multimodal knowledge points that were not involved, and obtained 784 knowledge points. Although the mathematical problems in "Zujuan" do not have knowledge point annotations that can be directly crawled and are not organized in the way of "Jiaoyan Cloud", we have conducted fine-grained annotation of knowledge points on the problem data from the "Zujuan" section. The math problem is first solved through GPT-4. The GPT-4 assisted annotation manual annotation details can be referred to Appendix E.4.

3.4 Comparison with Existing Benchmarks

The CMMaTH dataset is primarily used to evaluate multimodal math reasoning capabilities in K12 Chinese educational scenarios. We compared the current mainstream multimodal mathematical datasets and large model benchmarks in Table 1. Compared to existing multimodal benchmarks and multimodal reasoning benchmarks, the CMMaTH dataset has the following characteristics:

Tailored for real Chinese K12 Multimodal scenarios MathVista features a substantial number of problems that are associated with natural and synthetic images. However, these images do not accurately represent the genuine data distribution encountered in K12 mathematics educational settings. OlympiadBench is a bilingual multimodal benchmark at the Olympiad level, but it is too challenging and doesn't align well with real K12 multimodal math scenarios. Additionally, the variety of multimodal visual elements is relatively limited. Compared to comprehensive datasets like MMMU and CMMMU, CMMaTH is specifically

designed for Chinese subjects and exhibits significant diversity in multimodal Chinese math problems. Instead, We collect multimodal mathematical question data from real Chinese exam aggregation websites, specifically tailored to the Chinese K12 educational context. As illustrated in Figure 7, the questions in the CMMaTH dataset require comprehensive understanding multimodal elements, including Chinese text and math symbols, as well as mastery of K12 knowledge points.

High-quality Fine-grained Annotation and Evaluation Tool Every question in our dataset is meticulously annotated with standardized answers, solutions expressed in natural language, associated multimodal knowledge points, visual element categories, and K12 grade levels. This fine-grained annotation enables a more nuanced evaluation of multimodal mathematical proficiency within the K12 educational context. Despite the fact that numerous benchmarks, such as MathVista and GeoEval, rely on GPT-4 for answer extraction and validation, we introduce an open-source model named GradeGPT. GradeGPT stands out by providing a stable, cost-free, and swift accuracy evaluation specifically tailored for the CMMaTH dataset. Also, this approach makes it easier to dynamically maintain our dataset.

Extreme Diversity Currently, high-quality Chinese multimodal mathematics datasets are scarce. MATH-VISION lacks Chinese content, MATH-VISTA has only a few Chinese samples, and CMMU includes just 540 math problems, which are not detailed enough. We have included about 23k

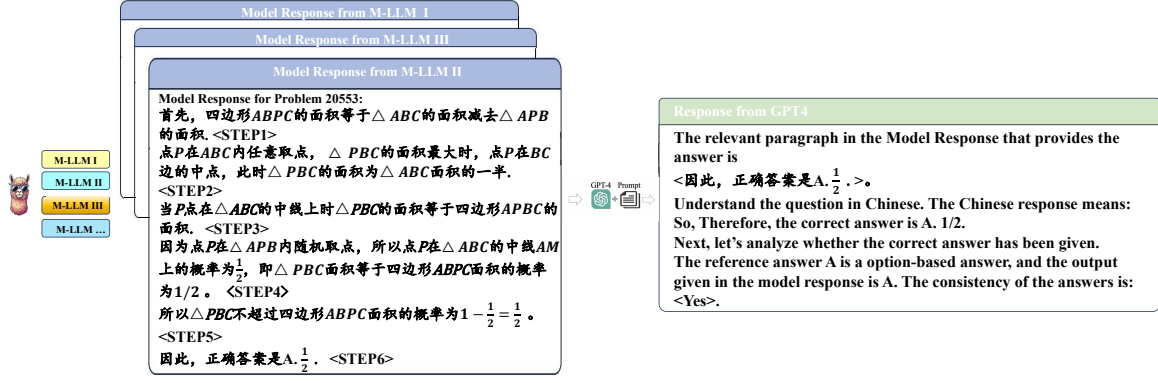


Figure 3: Instruction Construction Pipeline of GradeGPT.

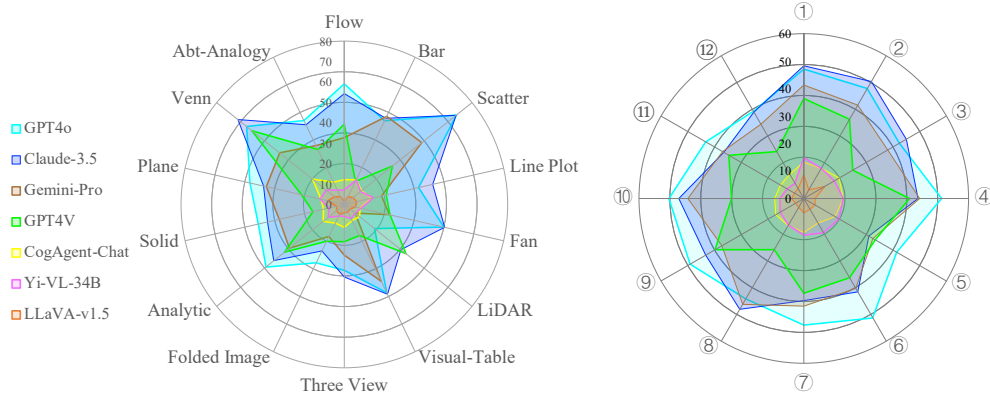


Figure 4: The results of mainstream multimodal large models and pure text large models on the CMMAth dataset. **Left:** represents the performance evaluation of selected LMMs and LLMs across various Visual Subjects. **Right:** the performance assessment of these models on different educational grade-level questions.

fine-grained multimodal mathematics assessment samples, covering 14 K12 mathematics visual categories, making it the largest known multimodal Chinese dataset to date. The CMMAth dataset also includes a diverse range of question types, featuring many multi-choice questions, free-form answer questions, and "Composition Questions," as illustrated in Table 2 and Figure 8, which are commonly found in real-world Chinese mathematics education.

4 GradeGPT

The CMMAth dataset encompasses a large variety of problem-solving objectives, such as mathematical expressions, multiple-choice options, numerical outcomes, coordinate points, conclusion figures, and correctness assessments. Traditionally, in reasoning or evaluation contexts, problems have been formulated as multiple-choice or true/false questions to facilitate comparison and to simplify the extraction of results. Also, it is difficult to maintain dynamically updated benchmark. Employing API

models for evaluation is prohibitively expensive, and the resulting evaluations are not consistently stable, which also hampers the iterative development of models on benchmarks, such as hyperparameter selection.

To provide a stable, free, fast, and easy-to-update model response evaluation tool, we introduce GradeGPT, an answer comparison model tailored for the CMMAth dataset. GradeGPT is designed to receive a question, its standard answers, and a model-generated response. It extracts key steps including results from Chinese output. Determine whether the result is consistent with the standard answer. Our GradeGPT is a streamlined, open-source model. When integrated with frameworks such as vLLM using the 14B model, it can swiftly compare a myriad of model-generated answers, accomplishing a remarkable judgment accuracy of 96.1% for assessing responses comparable with GPT-4 API.

Prompt Format

In the prompt input of GradeGPT, there are "ques-

Model	Overall	Flow	Bar	Scatter	Line Plot	Fan	LiDAR	Visual-Table	Three View	Folded Image	Analytic	Solid	Plane	Venn	Abt-Analogy
LLMs(Text-only, Zeroshot)															
Baichuan-13B(Yang et al., 2023)	8.4	6.7	4.8	12.2	12.4	13.1	16.2	5.4	4.1	8.5	11.1	6.7	13.7	12.8	9.3
Qwen-14B(Bai et al., 2023)	13.7	15.5	7.3	14.3	16.9	13.6	10.8	11.4	12.8	14.8	15.9	12.7	17.8	20.4	19.3
LLama2-70B(Touvron et al., 2023)	4.5	4.7	2.5	4.4	7.9	7.4	8.1	3.4	5.4	5.1	5.3	4.1	5.3	5.9	4.5
Math LLMs(Text Input, Zeroshot)															
MetaMath-70B(Yu et al., 2024)	5.7	4.6	3.3	6.6	8.7	5.7	0.2	4.2	4.1	8.5	7.2	4.8	8.5	9.8	5.4
DeepSeek-Math-7B(Shao et al., 2024)	14.0	13.4	6.7	14.7	13.1	12.5	12.2	8.1	13.5	12.3	17.2	16.5	21.6	19.5	13.8
InternLM2-Math-20B(Ying et al., 2024)	6.2	4.9	3.4	6.6	9.5	5.7	1.0	4.0	3.9	8.8	8.5	5.3	9.1	9.7	6.0
MAmmoTH2-8x7B(Yue et al., 2024)	16.1	14.1	8.0	15.3	13.4	13.2	12.7	9.4	14.4	12.8	18.0	17.3	21.6	20.3	14.5
Open-source LLMs (Text + Image, Zeroshot)															
LLaVA-v1.5-7B(Liu et al., 2023a)	5.5	1.5	4.2	5.4	6.2	5.4	3.6	4.0	4.2	5.3	4.8	3.9	8.4	6.1	4.2
LLaVA-NEXT-8B(Liu et al., 2024a)	5.4	7.0	2.1	6.8	5.9	1.7	8.1	4.4	6.7	4.2	6.2	4.2	7.7	7.5	6.6
LLaVA-OneVision-7B(Li et al., 2024)	13.7	12.8	5.2	11.4	10.9	9.7	8.1	8.1	13.8	16.5	16.8	13.6	25.2	19.3	13.6
Yi-VL-34B(Young et al., 2024)	8.3	7.1	4.6	10.2	14.6	8.5	6.8	7.7	5.9	6.4	10.1	7.8	12.2	11.3	7.9
CogVLM-18B-Chat(Wang et al., 2023b)	9.4	10.6	4.6	9.5	12.0	7.5	8.4	8.1	10.2	9.7	12.1	9.2	10.2	19.0	10.8
Qwen2-VL-7B(team, 2024)	10.0	10.3	3.5	10.4	6.5	13.6	1.4	7.5	10.2	11.0	10.6	6.8	20.0	13.0	11.8
CogAgent-18B-Chat(Hong et al., 2023)	10.6	12.2	5.2	10.8	13.7	8.0	9.5	8.8	11.2	10.2	13.2	10.5	11.8	19.9	12.2
InternLM-XComposer2-VL(Dong et al., 2024)	3.4	3.3	5.3	3.2	6.2	11.3	6.2	5.4	4.0	0.5	0.4	3.6	1.5	1.8	3.6
InternVL2-8B(Chen et al., 2024b)	23.9	44.6	11.0	22.4	22.6	29.0	14.9	17.4	22.3	14.4	25.0	19.4	30.5	25.1	19.7
Closed-source LLMs (Text + Image, Zeroshot)															
GPT4V(OpenAI, 2023)	27.0	39.3	12.5	30.2	21.0	22.9	38.6	16.9	18.3	20.0	37.5	15.8	21.5	58.0	29.9
Gemini-Pro(Anil et al., 2023)	35.7	58.9	21.1	47.1	31.2	50.6	32.4	27.4	24.7	20.3	37.3	29.5	42.2	53.8	31.5
Claude-3.5(Anthropic, 2024)	37.4	63.4	20.9	56.6	44.3	60.2	35.1	31.7	30.8	21.6	37.6	29.1	37.7	59.6	38.8
GPT4o(OpenAI, 2024)	47.8	59.1	45.5	70.3	37.3	50.0	18.9	48.0	32.2	31.8	49.1	43.2	47.4	61.2	45.6
Heuristics baselines															
Random Guess	14.4	13.3	7.9	15.4	13.5	12.4	12.8	7.9	13.2	12.3	17.2	16.4	21.8	19.6	14.3
Frequent Guess	15.1	14.6	7.6	16.0	14.5	13.8	13.5	9.2	15.2	13.8	17.8	17.3	22.6	20.1	14.3
Human Performance															
Human (testmini)	80.1	73.7	78.9	96.2	95.1	57.4	91.7	83.5	69.2	63.2	67.5	51.6	72.1	89.1	83.1

Table 3: Comparison of model performances across various mathematical visual subjects. Visual subjects: Flow: Flow Chart, Bar: Bar Chart, Scatter: Scatter Chart, Line Plot: Line Curve and Plot, Fan: Fan Chart, LiDAR: LiDAR Chart, Visual-Table: Visual-Table Chart, Three View: Three View Graph, Folded Image: Folded Image Graph, Analytic: Analytic Geometry Problem, Solid: Solid Geometry Problem, Plane: Plane Geometry Problem, SolG: Venn: Set Venn Graph, Abt-Analogy: Abstract Analogy Graph. The first and second highest accuracy of LLMs are marked in red and blue, respectively.

tions," "reference answers," and "model output answers." The model is required to provide an answer in the form of "<Yes>" or "<No>" indicating whether the model output answer is equivalent to the standard reference answer. We have designed an instruction format named Cross-Lingual-Judge-of-Chain(CL-JoC) for the purpose of determining answer consistency. CL-JoC first analyzes the model response and finds the key sentences that give the answer in the model response, understand key chinese sentences in English. Then analyze the standard answer, determine the type of the standard answer, and then determine whether the standard answer is included in the model response. More details can be found in Appendix G.

Instruction Construction

We first generate inference results on CMMaTH using multiple Multimodal LLMs and provide GPT-4 with a detailed few-shot prompt to synthesize answer judgments in the form of a Cross-Lingual Judge-of-Chain response. By employing GPT-4's In-Context Learning, as shown in Figure 3, we have established a procedure for synthesizing instruction data and have produced approximately 56k cross-lingual result judge instruction pairs. Through fine-tuning the model with these instructions, we obtained an expert model, GradeGPT, which possesses the capability to compare answers.

5 Experiments

We conducted a comprehensive series of experiments to evaluate various models on the CM-MaTH dataset. Specifically, we assessed multiple LLM/LMMs, including 15 open-source models and 4 API-based closed-source models. We also evaluated the performance variations of different models under conditions augmented with auxiliary information, such as OCR Caption result. Additionally, we investigated the effectiveness of cross-lingual reasoning techniques in enhancing the multimodal mathematical capabilities of LMMs in the Chinese multimodal context. Through systematic experimental design and data analysis, our objective was to elucidate the strengths and weaknesses of these models in handling complex Chinese multimodal contexts. Further details on experiments related to the dependency on visual elements within the CMMaTH dataset, as well as the implementation of contextual learning capabilities in LMMs and evaluation details can be found in Appendix 6.

We also conducted a detailed analysis and evaluation of GPT4o on a random miniset CMMaTH, categorizing errors into four types: Perceptual Errors, Reasoning errors, Calculation errors, and Reject Errors. The error type distribution of GPT-4o on CMMaTH is shown in Figure 6. A more detailed definition of hallucination types can be found in

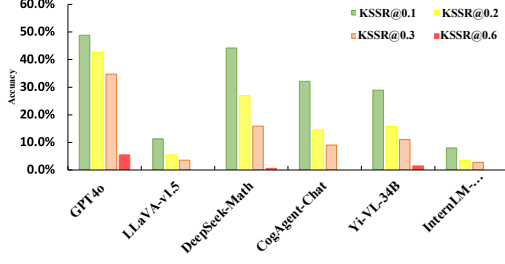


Figure 5: The metrics of different LMM/LLM models about KSSR.

Appendix F.

5.1 Main Experiments on LLM/LMMs

We evaluated the results of mainstream multimodal large models and mathematical expert models in Table 3. We analyzed the trend of existing large models in descending with problems and conditions, as well as the effectiveness of techniques such as Cross-Lingual Prompting in solving Chinese multimodal mathematical problems. The experimental results indicate that our data exhibits extremely strong diversity and relatively challenging reasoning depth. Figure 4 and Table 3 show models such as GPT-4o struggle to comprehend our multimodal content and reasoning questions effectively, resulting in significant performance gaps between open-source and proprietary models.

5.2 Knowledge Skill Analysis

Quantitative Analysis We have formulated a **Knowledge Successful Solve Rate(KSSR)** as a structural metric to gauge the proficiency level of multi-modal extensive models in mastering knowledge points. N_{kn} is the total number of knowledge points of CMMaTH. Acc_{kn_i} is the proportion of correct answers to questions labeled as i 'th knowledge point. I denotes an indicator function.

$$KSSR@α = \frac{\sum_{i=1}^{N_{kn}} I(Acc_{kn_i} > α)}{N_{kn}} \quad (1)$$

We contend that a knowledge point can be deemed comprehensively understood only when the accuracy rate of solving problems related to that knowledge point surpasses a predefined threshold, denoted as $α$. For our investigation, we have established $α$ at the values of 0.1, 0.2, 0.3, and 0.6 to demarcate the levels of mastery. As shown in Figure 5, our experiment showed that when subjected to a more stringent KSSR metric standard, the most advanced models performed poorly.

Qualitative Analysis Based on the fine-grained

LMM	Overall-Acc
LLaVA-v1.5	5.5
InternLM-XComposer2-VL	3.4
Gemini-1.5-Pro	37.2
LLaVA-v1.5 + En-CoT	9.4(+3.9)
InternLM-XComposer2-VL + En-CoT	16.9(+13.5)
Gemini-1.5-Pro + En-CoT	41.1(+3.9)
LLaVA-v1.5 + CLP	12.7(+7.2)
InternLM-XComposer2-VL + CLP	17.1(+13.7)
Gemini-1.5-Pro + CLP	43.8(+6.6)

Table 4: The performance of train-free CoT reasoning techniques on the CMMaTH dataset.

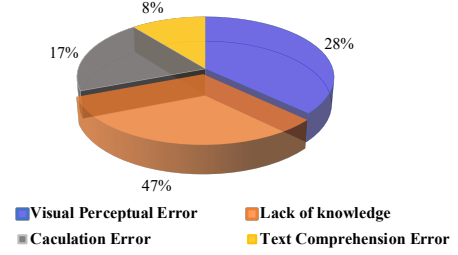


Figure 6: Distribution of Error Types in GPT-4o.

knowledge point annotation, we conducted a detailed knowledge point skill level analysis of current LMMs. We ranked the knowledge points based on their respective problem-solving rates in Appendix B. Our analysis revealed the significant gap between commercial models and open-source models, as well as the distribution of the multimodal knowledge points mastered.

5.3 Experiments of Cross-language Reason Technology

We also attempted several multilingual Chain-of-Thought approaches such as En-CoT, CLP(Cross-Lingual Prompting) used by Qin et al. (2023) to observe whether multimodal mathematical problems could be enhanced through context learning techniques without training. The results indicate that multilingual CoT methods face challenges in solving, possibly due to the abundance of Chinese contextual text in the image content, which may necessitate the model to demonstrate excellent cross-lingual OCR capabilities. We have included more details on the implementation of Cross-Lingual Prompting and En-CoT on the CMMaTH dataset in Table 4. The prompt implementation detail can refer to Appendix H.3.

6 Conclusions

We introduce CMMaTH, a detailed Chinese K12 multimodal reasoning benchmark with diverse question types, visual elements, and complex rea-

son depth. The benchmark includes detailed knowledge points, standard solutions, and grade levels to measure the mastery of knowledge points in the K12 multimodal math skill. To evaluate large multimodal models quickly and affordably, we built GradeGPT, an open-source tool for assessing results. Extensive experimental results on CMMaTH manifest the limitations of current models.

Limitation & Potential Impact

Our dataset CMMaTH, as a multimodal mathematics dataset aimed at the K12 education sector, can facilitate model evaluation and iteration of multimodal large models in this field, and may promote the research and development of educational artificial intelligence. GradeGPT is a result-oriented, relatively coarse reasoning response evaluator. How to construct a process evaluation model for fine-grained assessment of the reasoning ability of large models remains an area to be explored in the future.

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A More Related Work

A.1 Multimodal Large Model Evaluation

The multimodal large models face serious hallucination issues in perceiving objects and executing reason and inference (Zhang et al., 2024a; Mei et al., 2024a; Jin et al., 2024b,a; Shi et al., 2023b). How to evaluate and reduce inference hallucinations of MLLMs (Chen et al., 2024a; Bi et al., 2024a,b) has received widespread attention. To systematically evaluate the various capabilities of multimodal large models, diverse multimodal benchmarks are utilized for assessing the abilities of large models and aiding iterative development. POPE (Li et al., 2023b) is used to evaluate the accuracy of large models in identifying perceptual objects. MMMU and CMMMU (Yue et al., 2023; Zhang et al., 2024b) are comprehensive subject datasets design to assess the proficiency of large models in mastering massive multimodal multi-disciplinary knowledge. SEED-Bench designed 19,000 diverse multimodal questions spanning video and image modalities to evaluate the spatiotemporal capabilities of multimodal large models (Li et al., 2023a). MMVet (Yu et al., 2023) attempts to design datasets to evaluate the integrated capabilities of different multimodal large model systems in combining various Vision-Language skills.

A.2 Concurrent Work Discussion

MathBench (Liu et al., 2024b) is the first attempt to introduce a fine-grained knowledge point system (Huang et al., 2024a; Bi et al., 2024d) for evaluating large models’ mathematical abilities. However, MathBench is a purely text-based mathematical benchmark and does not involve multimodal skills, such as understanding diagrams, and its annotation granularity is relatively coarse.

MathScape (Zhou et al., 2024a) and CMM-Math (Liu et al., 2024c) are concurrent works that were developed after ours, but MathScape only contains 1/10 of the data size and a coarser knowledge point system. MathScape also adopts a method that embeds problem text into images for evaluation, focusing more on end-to-end recognition and solving of problems. CMM-Math provides a dataset for training and evaluation, with around 5k test samples for assessment. In contrast, our CMMaTH dataset is larger in scale, with finer annotation granularity, and is specifically designed for evaluating multimodal mathematical capabilities in Chinese.

B Analysis of K12 knowledge point mastery level of LMMs

In addition to KSSR, based on the fine-grained knowledge point annotation of CMMaTH, we also clustered several LMMs on the specific level of mastering K12 knowledge points. According to the accuracy of solving the questions corresponding to each knowledge point, for GPT-4o, Gemini-Pro and Claude-3.5, LLaVA-v1.5, we listed the top 20 knowledge points that these models have the best in Figure 16, Figure 17, Figure 18, Figure 19.

At the same time, we identified the K12 knowledge points that these models struggle with and visualized them in Figure 20. Our fine-grained annotation and analysis reveal the shortcomings of current LMMs in solving K12 education problems from the perspective of knowledge points.

Gemini-Pro appears to be optimized for large-scale triangle-related geometry problems but lacks the balanced knowledge and skill level found in models like Claude-3.5 and GPT-4. Additionally, our analysis suggests that open-source models generally struggle with fundamental knowledge skills such as parallelism and positional relationships. We hope that our datasets and tools can promote the data collection and synthesis of existing LMMs from a perspective based on knowledge points.

C Analysis About Visual/Auxiliary Information for LLMs Inference on CMMaTH

C.1 The Impact of OCR Information

OCR information includes important information such as Chinese characters on the coordinate axes in mathematical abstract forms, recognized mathematical symbols, etc., and plays a key role in assisting the understanding of visual information. We also evaluated the ability of OCR information for LLM/LMMs to solve CMMaTH Chinese multimodal mathematics questions.

C.2 The Impact of Visual Content on LMM

In real questions in the real world and academic datasets like MathVista, the text part of many questions contains descriptions of the visual part, which leads to many models that may be able to solve mathematical problems with text reasoning capabilities. In order to evaluate our dataset Regarding the degree of dependence on the visual part, we evaluated the problem-solving capabilities of relatively

Model	Overall	Flow	Bar	Scatter	Line Plot	Fan	LiDAR	Visual-Table	Three View	Folded Image	Analytic	Solid	Plane	Venn	Abt-Analogy
LMMs(Text+Image+OCR Caption, Zeroshot)															
LLaVA-v1.5-7B	5.5	1.5	4.2	5.4	6.2	5.4	3.6	4.0	4.2	5.3	4.8	3.9	8.4	6.1	4.2
Yi-VL-34B	8.3	7.1	4.6	10.2	14.6	8.5	6.8	7.7	5.9	6.4	10.1	7.8	12.2	11.3	7.9
Qwen2-VL-7B	13.7	15.5	7.3	14.3	16.9	13.6	10.8	11.4	12.8	14.8	15.9	12.7	17.8	20.4	19.3
LLaVA-v1.5-7B+OCR Caption	4.9	5.2	1.3	3.4	4.5	5.9	4.4	2.6	3.5	3.9	4.6	3.8	3.7	8.0	5.6
Yi-VL-34B + OCR Caption	10.2	8.3	5.6	11.6	14.8	9.8	7.4	9.2	5.9	6.8	10.8	8.4	13.2	13.0	9.4
Qwen2-VL-7B + OCR Caption	14.1	16.1	8.5	15.1	18.2	14.1	12.3	12.6	13.6	16.3	17.0	14.1	19.3	21.2	20.5
GPT4V	27.0	39.3	12.5	30.2	21.0	22.9	38.6	16.9	18.3	20.0	37.5	15.8	21.5	58.0	29.9
Gemini-Pro	35.7	58.9	21.1	47.1	31.2	50.6	32.4	27.4	24.7	20.3	37.3	29.5	42.2	53.8	31.5
Claude-3.5	37.4	63.4	20.9	56.6	44.3	60.2	35.1	31.7	30.8	21.6	37.6	29.1	37.7	59.6	38.8
GPT4o	47.8	59.1	45.5	70.3	37.3	50.0	18.9	48.0	32.2	31.8	49.1	43.2	47.4	61.2	45.6
Gemini-Pro + OCR Caption	39.7	34.5	50.0	50.9	21.0	25.3	8.5	44.4	26.7	19.5	36.7	35.2	41.0	42.7	33.8
GPT4o + OCR Caption	50.1	61.5	47.7	72.9	39.7	52.6	21.2	50.1	34.8	34.3	51.5	45.0	50.0	63.7	48.0
Claude-3.5 + OCR Caption	46.7	55.6	48.0	70.9	45.9	51.5	36.2	49.8	36.5	25.9	45.1	39.7	41.8	67.8	44.2
LMMs(Text-only, Zeroshot)															
GPT4o-w/o Visual Diagram	17.9	45.0	7.0	39.2	17.4	45.8	19.5	18.6	17.7	10.1	28.4	19.8	19.1	43.2	17.5
Gemini-Pro-w/o Visual Diagram	14.8	25.0	4.3	18.0	7.6	24.4	12.9	14.2	11.3	1.4	18.2	4.9	18.8	31.0	10.0
Claude-3.5-w/o Visual Diagram	19.9	37.0	10.1	32.0	18.1	46.8	17.2	13.4	20.0	7.5	29.8	15.1	20.4	42.2	15.6

Table 5: Model performance of LLMs, Math LLMs, and various commercial API models on CMMaTH with auxiliary OCR caption input. The results of the advanced model removing visual information are also reported in the table. The definitions of different subjects are the same as in Table 3. The **first** and **second** highest accuracy of LMMs are marked in red and blue, respectively.

strong interface models, such as GPT-4o, Gemini-Pro, and Claude-3.5, when there is no visual input and only text input. Our results in Table 5 show that GPT-4o and Gemini-Pro and Claude-3.5 suffer huge performance degradation in the absence of visual input. This shows that in addition to understanding the text part, a large number of questions in our benchmark require a full understanding of the corresponding visual elements in order to solve the questions.

C.3 Few-Shot Evaluation on CMMaTH Dataset

We evaluated the reasoning capabilities of several advanced API-based models under three few-shot conditions(3-shot) in Table 8. The results indicate that few-shot learning can slightly enhance the performance of current commercial-grade API models. However, there remains a significant gap in effectively addressing multimodal mathematical problems in real-world Chinese multimodal reason contexts compared to our human evaluation baseline.

D Model Generation Details

D.1 Model Weight Version

We have listed the parameter versions and the Hugging Face repository names of the open-source models and version for API-based model used in Table 16.

D.2 Model Sampling Parameter

We have listed the corresponding hyperparameters used by the models in Table 15. For API models, we have indicated the corresponding release

Statistic	Number
Single choice's distribution of question options	
- Proportion of answers A	2694(14.8%)
- Proportion of answers B	3903(21.4%)
- Proportion of answers C	3961(21.7%)
- Proportion of answers D	3148(17.5%)
Knowledge point number	784
Levels	5
Visual Subjects	13
Maximum question length	593
Minimum question length	6
Average question length	75.1
Grade Distribution Elementary(1-6)	800
Junior(7-9)	5082
Senior(10-12)	17972

Table 6: Key statistics of CMMaTH. The unit of question length is words.

versions. Models using vLLM for inference are annotated.

E CMMaTH Dataset Details

E.1 More Detailed Statistics

For readers' reference and understanding of the characteristics of the dataset, we provide other statistical information on our CMMaTH dataset, statistics on options, number of knowledge points, question difficulty level, number of visual element types in Table 6. We have also provided representative examples in Figure 7 and new question type example "Composition Question" from the dataset in Figure 8.

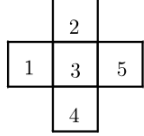
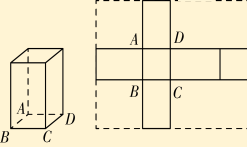
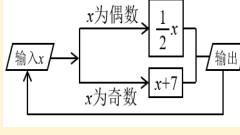
Abstract Analog Graph	Folded Image Graph	Flow chart	Stem-and-Leaf Display												
Question: 如图所示, 现有一迷失方向的小青蛙在3处, 它每跳动一次可以等可能地进入相邻的任意一格 (若它在5处, 跳动一次, 只能进入3处, 若在3处, 则跳动一次可以等机会地进入1, 2, 4, 5处), 则它在第三次跳动后首次进入5处的概率是 () A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{3}{16}$ D. $\frac{1}{6}$ 	Question: 如图所示, 图2中实线围成的部分是长方体 (图1) 的平面展开图, 其中四边形ABCD是边长为1的正方形, 若向虚线围成的矩形内任意抛掷一质点, 它落在长方体的平面展开图内的概率是$\frac{1}{4}$, 则此长方体的体积是_____。 	Question: 如下图所示, 这是一个流程图, 在图示的运算程序中, 若开始输入的x的值为100, 我们发现第1次输出的结果为50, 第2次输出的结果为25, 第三次的输出结果为33, 第四次输出结果为40..., 则第2021次输出的结果为_____。 	Question: 如图所示的茎叶图记录了甲、乙两组各5名工人某日的产量数据 (单位: 件). 若这两组数据的中位数相等, 且平均值也相等, 那么x和y的值分别为 () A.3,5 B.5,5 C.3,7 D.5,7 <table data-bbox="1075 456 1315 613"><tr><th>甲组</th><th></th><th>乙组</th></tr><tr><td>6</td><td>5</td><td>9</td></tr><tr><td>2 5</td><td>6</td><td>1 7 y</td></tr><tr><td>x 4</td><td>7</td><td>8</td></tr></table>	甲组		乙组	6	5	9	2 5	6	1 7 y	x 4	7	8
甲组		乙组													
6	5	9													
2 5	6	1 7 y													
x 4	7	8													
Answer: C Level: High	Answer: 3 Level: High	Answer: 8 Level: Middle	Answer: A Level: Primary												
Knowledge point: 独立事件的概率乘法计算 Probability multiplication of independent events calculate	Knowledge point: 计算面积体积相关的几何概率 Calculate geometric probabilities related to area and volume	Knowledge point: 规律探索及程序框图 Law exploration and program block diagram	Knowledge point: 使用样本估计整体、茎叶图 Use samples to estimate the population, stem and leaf diagrams												

Figure 7: Some samples related to "Folded Image Graph", "Flow chart", "Stem-and-Leaf Display" visual subject on the CMMaTH dataset

E.2 Data Source Detail

These two electronic data websites, "Jiaoyan Yun"³ and "Zujuan"⁴, mainly collect electronic exam paper data from primary, middle, and high schools. The electronic test paper data includes detailed information on the grade level of the exam, the exam date, the school administering the exam, and the exam scope. The Jiaoyan Cloud, containing more than 1.2 million Chinese math problems, forms a major part of our private database. Additionally, we have crawled a large number of multimodal math test questions from "Zujuan" totaling around 400k multimodal math questions. We did not crawl all the data, especially since crawling data from Jiaoyan Cloud requires obtaining the corresponding copyrights. Ultimately, the ratio of data in our private database from Jiaoyan Cloud to Zujuan is approximately 7:3.

To format all questions for use, we processed them by OCR engine like Mathpix⁵ interface. Due to inherent errors in the OCR engine, we introduced manual checks to ensure the accuracy of parsing results and to verify whether the questions belong to multimodal math problems.

To more clearly elucidate our data collection process, we have depicted the overall pipeline of

data collection in Figure 15.

E.3 Quality Check Detail

Since we use GPT-4 as a quality check tool, we provide the prompt for GPT-4 quality check in Figure 9, GPT-4 assigned a confidence level to the data on a scale of 1 to 5. Data with a confidence level lower than 3 were filtered out. To ensure the high quality of the final data, we conducted sampling and manual verification. We performed three random samples, each consisting of 500 multimodal samples, to check the data quality and ensure the consistency of the knowledge points and data. When verifying whether problem is solvable, we use multiple closed-source interface models (GPT-4o, Gemini-Pro and Claude-3.5-Sonnet) to solve each problem. For problems that cannot be solved, we perform manual checks to compare whether the reference analysis given is correct and whether the problem can be solved.

E.4 Knowledge Point Assistant Labeling Detail

The data from Jiaoyan Yun already includes detailed knowledge point classifications. We have also provided fine-grained knowledge point annotations for the questions sourced from Zujuan. The GPT-4 prompt for knowledge point labeling includes a detailed documentation of knowledge

³<https://www.jiaoyanyun.com/>

⁴<https://zujuan.xkw.com/>

⁵<https://mathpix.com/>

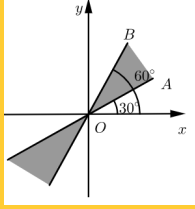
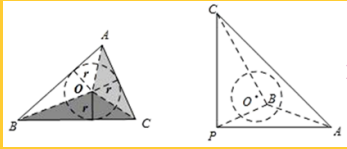
Plane Geometry(Middle)	Solid Geometry(High)
Question: 如图, 分别写出适合下列条件的角的集合。 (1) 终边落在射线OB上: _____。 (2) 终边落在直线OA上: _____。 (3) 终边落在阴影区域内(含边界): _____。 	Question: 设O是$\triangle ABC$的内切圆圆心, $r_{\text{内}}$是$\triangle ABC$的内切圆半径, 设S是$\triangle ABC$的面积, l_{ABC}是$\triangle ABC$的周长。 (1) 与三棱锥的四个面都相切的球叫做三棱锥的内切球。设三棱锥的体积是V, 表面积是S, 请用类比推理思想, 写出三棱锥的内切球的半径公式$R_{\text{内}}$ (2) 如图2, 在三棱锥$P-ABC$中, PA、PB、PC两两垂直, 且$PA = PB = PC = 1$, 求三棱锥$P-ABC$的内切球半径和外接球的半径之比。 
Answer: (1) $\{\alpha \alpha = 60^\circ + k \cdot 180^\circ, k \in \mathbb{Z}\}$ (2) $\{\alpha \alpha = 30^\circ + k \cdot 180^\circ, k \in \mathbb{Z}\}$ (3) $\{\alpha 30^\circ + k \cdot 180^\circ \leq \alpha \leq 60^\circ + k \cdot 180^\circ, k \in \mathbb{Z}\}$	Answer: (1) $R_{\text{内}} = \frac{3V}{S}$ (2) $\frac{\sqrt{3}-1}{3}$
Knowledge point: 根据终边位置描述角的范围 Describes the extent of an angle in terms of its terminal edge position	Knowledge point: 合情推理与演绎推理 Plausible reasoning and deductive reasoning

Figure 8: Examples of Composition questions question types

GPT-4 Data Quality Check Prompt Example: Role: System You are a data quality scoring assistant. Please refer to the scoring criteria and provide a reference score for a Chinese math Question. The scoring criteria for reference are as follows: 1: This question has serious grammar issues due to the lack of coherence between the preceding and following sentences. 2: The sentences in this question are generally smooth, but there are formatting issues with the LaTeX formula and table. 3: The sentences in this question are generally smooth, and the format of the LaTeX formula and table is basically correct. However, there may be some missing conditions, redundant noise text, or contradictory conditions. 4: The sentence of this question is smooth, the problem description is basically clear, the LaTeX formula is correct, and the conditions given in the question are not contradictory. 5: The sentence of this question is fluent, the problem description is basically clear, the LaTeX formula is expressed correctly, the conditions given in the question are not contradictory, and it is preliminarily estimated that they can be solved. Role: User. Math Question: {question} Estimated Score:
--

Figure 9: Prompt used for quality inspection with GPT-4.

point types. Considering GPT-4's context length limitation and the large number of knowledge points, we assess whether a problem belongs to a specific category in batches. A GPT-4 Knowledge point classification prompt in Figure 10.

We used GPT-4 to formalize the standard form of knowledge points as a classification problem. After meticulous prompt processing, GPT-4 achieved an accuracy rate of 92%(We validated the preliminary accuracy of knowledge point annotation using GPT on a relatively 500 test set with only a few hundred questions). But this still did not meet our requirements. Therefore, GPT-4 is used here solely to assist with annotation and speed up the process. Afterward, manual verification is performed to en-

sure the accuracy of the annotations.

E.5 Visual Subjects Detail

We provide the Chinese and English explanations of the Visual Subjects involved in the topics in the dataset in Table 7.

The definition of "Visual Subject" was extracted and screened from the knowledge point names of "Jiaoyan Yun". The classification of "Visual Subject" is coarser compared to the Jiaoyan Yun system and is based solely on the types of images used in multimodal math problems. Jiaoyan Yun's question knowledge point annotations involve the examination of specific "Visual Subjects." For example, for the "Visual-Table" subject, it involves Jiaoyan

Prompt Example:**Role: System**

You are a math problem analysis assistant. Please determine whether the knowledge points tested by a math problem belong to a series of knowledge points based on the definition of the corresponding knowledge points, If there are no matching knowledge points here, provide output <No_Match>:

List of reference knowledge points are:

Chinese Knowledge Point 1: “根据折线图计算中位数” which means Calculate the median based on the line chart

Chinese Knowledge Point 2: “等腰三角形外角的性质：三角形的一个外角等于与其相邻的两个内角之和” which means: The properties of the outer angle of an isosceles triangle: One outer angle of a triangle is equal to the sum of two inner angles that are not adjacent to it.

Chinese Knowledge Point 3:

.....

Chinese Knowledge Point 10:

For Example:

Example 1:

Math Question: 如图， AD 是 $\triangle ABC$ 的 BC 边上的高， AE 平分 $\angle BAC$ ，若 $\angle B=42^\circ$ ， $\angle C=70^\circ$ ，则 $\angle AEC$ 的度数是 。

Question Analysis:

$\angle B=42^\circ$ ， $\angle C=70^\circ$ ， $\angle BAC=180^\circ - \angle B - \angle C=68^\circ$ ， $\therefore AE$ 是角平分线， $\angle EAC=\frac{1}{2}\angle BAC=34^\circ$ 。 $\therefore AD$ 是高， $\angle C=70^\circ$ ， $\angle DAC=90^\circ - \angle C=20^\circ$ ， $\angle EAD=\angle EAC - \angle DAC=34^\circ - 20^\circ=14^\circ$ ， $\angle AEC=90^\circ - 14^\circ=76^\circ$ 。

Corresponding Knowledge Point: 等腰三角形外角的性质：三角形的一个外角等于与其相邻的两个内角之和

....

Role: User

Math Question: {question}

Corresponding Knowledge Point:

Figure 10: Prompt form used to annotate knowledge points.

Yun’s knowledge point "calculation of median and mode based on statistical tables." For "Solid Geometry", it involves "volume calculation of solid geometric shapes".

E.6 Knowledge Point Detail

The CMMaTH dataset mainly adopts the knowledge point system of "Jiaoyan Yun". "Jiaoyan Yun" is a relatively mature commercial knowledge graph widely used in the field of Chinese mathematics education. It has undergone long-term user validation in the mathematics education sector. Compared to the coarse-grained knowledge point classification structures used by MathBench and MM-Math, "Jiaoyan Yun" employs a very rich knowledge structure with a massive scale of knowledge points, which better meets the needs of real-world educational scenarios in the industry.

MathBench(Liu et al., 2024b) is another related work trying to provide knowledge point label in math evaluation benchmark. However, their knowledge system is build by "Subject Area" and "Topic", which has coarse-grained classification. We provide some comparisons of some knowledge points here and the one-to-many correspondence between them in Figure 12 and Figure 13.

We provided detailed annotations of knowledge points for our dataset and conducted preliminary clustering of these knowledge points in "Jiaoyan Yun". The distribution of knowledge points in different clusters is showed in Figure 11.

E.7 Question Level Detail

The "Level" is a reference question difficulty provided by the Teaching and Research Cloud, which can be obtained through OCR tags during crawling. It is manually marked by teachers and corresponds to difficulty levels 1-5: "very simple", "simple", "moderate difficulty", "relatively difficult", "difficult".

E.8 Characteristics Of Annotators

We utilized a standard team of eight people, who spent two weeks annotating the data. All annotators have a university undergraduate education and are well-versed in basic knowledge of the K12 education field. To ensure quality, each question was verified by at least two people.

E.9 Heuristics baselines Detail

Similar to MathVista, we added two heuristic Baselines. These two heuristic strategies can only han-

MathBench 1 type		
★ 高中-中级几何-立体几何	High--Intermediate Geometry--Solid Geometry	
MathScape 4 types		
★ 空间向量与立体几何点之间的位置关系	★ 空间向量与立体几何空间几何体	★ 空间向量与立体几何综合
<i>The positional relation between space vectors and solid geometric points</i>	<i>Space vectors and solid geometry space geometry</i>	<i>Space vector and solid geometry synthesis</i>
CMMaTH 29 types		
★几何三视图求面积体积	★空间几何体的认识与理解	★空间向量与空间坐标系
<i>Calculate the area and volume of geometric three views</i>	<i>Knowledge and understanding of spatial geometry</i>	<i>Space vectors and space coordinate systems</i>
• 求常见几何体的表面积	• 棱柱、棱锥、棱台理解识别	• 旋转得到的立体图形
<i>Calculate the surface area of common geometries</i>	<i>Understanding and identifying prisms, pyramids, and pyramids</i>	<i>Rotated three-dimensional graphics</i>
• 球的截面问题	• 正方体对邻面展开图	• 空间向量数量积及坐标表示
<i>Ball cross section problem</i>	<i>Expanded of adjacent faces of cube</i>	<i>The quantity product of space vectors and coordinate representation</i>
•	•	•
• 旋转得到的立体图形	• 旋转得到的立体图形	• 建立空间直角坐标系
<i>Rotated three-dimensional graphics</i>	<i>Rotated three-dimensional graphics</i>	<i>Establish a spatial rectangular coordinate system</i>

Figure 12: Knowledge Point Annotation Comparisons in Topic Plane Circle.

Model	Overall	Flow	Bar	Scatter	Line Plot	Fan	LiDAR	Visual-Table	Three View	Folded Image	Analytic	Solid	Plane	Venn	Abt-Analogy
LMMs(Text+Image, Zeroshot)															
GPT4V	27.0	39.3	12.5	30.2	21.0	22.9	38.6	16.9	18.3	20.0	37.5	15.8	21.5	58.0	29.9
Gemini-Pro	35.7	58.9	21.1	47.1	31.2	50.6	32.4	27.4	24.7	20.3	37.3	29.5	42.2	53.8	31.5
Claude-3.5	37.4	63.4	20.9	56.6	44.3	60.2	35.1	31.7	30.8	21.6	37.6	29.1	37.7	59.6	38.8
GPT4o	47.8	59.1	45.5	70.3	37.3	50.0	18.9	48.0	32.2	31.8	49.1	43.2	47.4	61.2	45.6
LMMs(Text+Image, Few shot)															
Gemini-Pro(3-Shot)	39.3	34.9	49.7	50.5	20.8	24.9	9.0	44.1	26.6	19.7	36.5	34.6	41.0	42.5	34.0
Claude-3.5(3-Shot)	48.8	57.9	50.4	73.8	47.9	53.5	38.6	52.0	38.6	28.8	47.8	42.6	43.8	70.2	47.2
GPT4o(3-Shot)	52.2	63.9	50.4	75.3	42.9	55.0	24.0	53.6	37.0	36.8	54.3	48.6	52.6	66.1	50.5

Table 8: Model performance of few-shot(3-shot) experiment on CMMaTH.

knowledge for inference. The frequency of Reasoning Errors reflects the model’s logical and mathematical reasoning capabilities.

Calculation Errors

Calculation Error refers to the model performing incorrect mathematical operations, such as writing equations or solving equations incorrectly.

Reject Errors

Reject Error refers to the model’s inability to solve a problem that is actually solvable. The frequency of such errors reflects the model’s ability to follow instructions.

F.1 Case Study

We conducted a fine-grained manual evaluation of GPT-4V’s output on CMMaTH, with the results shown in Figure 21 to Figure 31.

G GradeGPT details

G.1 GradeGPT Prompt Detail

We have listed detailed Fewshot Examples using the GPT-4-generated GradeGPT model responses in Table 14. Through this table, you can observe the specific form of the Cross-Lingual-Judge-of-Chain that we have used.

G.2 GradeGPT Performance Metric

GradeGPT’s performance evaluation metric is precision in comparison. We constructed a model that responds to a test set containing outputs from various large models (including both correct and incorrect model outputs). Each output is labeled as correct or incorrect based on its result. GradeGPT is tasked with assessing whether the model responses are correct or incorrect, and this performance evaluation metric is a binary classification metric.

MathBench 3 points		
★ 高中-初级几何-立体几何	Middle--Basic Geometry--Basic Concepts of Geometry	
★ 初中-初级几何-圆的概念	Middle--Basic Geometry--Basic Concepts of Circles	
★ 高中-中级几何-直线和圆的方程	High--Intermediate Geometry--Equations of Lines and Circles	
MathScape 3 points		
★ 平面解析几何圆与方程 Plane analytical geometry circles and equations	★ 圆的图形性质 Graphic properties of circles	★ 竞赛知识点平面几何 Competition knowledge point of plane geometry
CMMaTH 49 points		
★与圆有关的计算 Calculations related to circles	★与圆有关的位置关系 Positional relation related to circles	★圆与其他数学概念 Circles and other concepts
• 圆的弦长问题 Questions about the chord length of a circle • 圆的标准方程 Standard equation of circle • • 圆的一般方程问题 General equation problems of circles	• 圆与多边形 Circles and polygons • 三角形内切圆 Triangle inscribed circle • 直线与圆 Lines and circles • • 圆的切线的相关问题 Questions about tangents to circles	• 圆与相似 Circle and similarity • 圆内接四边形的性质定理 Theorems about the properties of a quadrilateral inscribed in a circle • • 圆与平面直角坐标系 Circle and plane rectangular coordinate system

Figure 13: Knowledge Point Annotation Comparisons in Topic Solid Geometry.

G.3 GradeGPT Training Details

We generated cross-lingual evaluation instruction pairs using the outputs from InternLM-XComposer, LLaVA-v1.5, CogAgent-18B and Yi-VL-34B. These outputs were produced using GPT-4 Fewshot. The generated evaluation instructions were filtered based on specific rules, retaining only those responses from GPT-4 that contained the fields: <Yes>/<No>. Ultimately, we constructed a cross-lingual format instruction set comprising 56k instruction pairs.

GradeGPT was trained on 8 H800, with the Qwen-14B-Chat version used as the base model. The model’s batch size was set to 16. The learning rate was set to 1e-4, and the gradient accumulation step was set to 16. It was trained for 10 epochs on a 40k bilingual Judge-of-Chain dataset. A detail example of instruction can refer to Figure 14.

G.4 Futher More Ablation Study

We conducted experiments on a development set comprising outputs from a 0.5k model. The development set was sampled from a subset of 0.5k questions on CMMaTH. Each question was accompanied by answers provided by GPT-4V, GPT-4o, and middle school students. Each answer was manually annotated to indicate whether it was correct. We use 2 to measure the

answer judgment capability of different LMMs, including Zershot LMMs and LLMs after Finetune.

Ablation On Instruction Format We conducted experiments on various instruction enhancement techniques used by GradeGPT and compared the results with GPT-4 in Table 9. The results suggest that after various instruction enhancements, the accuracy of GradeGPT in model response judgment on CMMaTH can be improved to 96.1%, significantly surpassing the accuracy of GPT-4. The proposed strategy can significantly enhance GradeGPT’s ability to judge results. It is only slightly weaker than the performance of GPT-4(Fewshot) executed with a large number of examples. Our GradeGPT, as an open-source parameter model of approximately 14B, can serve as a stable, low-cost, and efficient alternative to GPT-4.

The Baseline we compared, Qwen-7B/14B(4-Shot), GPT-4(4-Shot), *Naive Outcome Finetune*, *Judge-of-Chain*. In the *Naive Outcome Finetune* format of instructions, the model is required to output its results indicating whether they are correct in the form of "<Yes>"/"<No>". *Judge-of-Chain* also includes the understanding of results and natural language descriptions of model outputs, but does not include the part of extracting key Chinese out-

Instruction Example:**Role: System**

You are an answer evaluation assistant. Based on the following questions and reference answers, determine if the model's output is the same answer. When making the determination, please provide the analysis process. Note: for multiple-choice questions, it is only correct if the output includes all the corresponding options. Note: if the model response does not specify a particular option, and its correctness cannot be determined, the answer consistency should be: <No>.

Role: User

以下是某道题目，答案和一条模型响应。请确定模型响应中是否给出了正确的参考答案。

Question: 如图：若 $\triangle ABE \cong \triangle ACD$ ， $\angle A = 50^\circ$ ， $\angle C = 20^\circ$ ，则 $\angle ADC = ?$ 。

Reference Answer: 110°

Model Response: 首先，我们需要了解这个问题的背景和目标。题目给出了三角形的三个角度，并要求我们找到另一个角度。我们可以通过三角形的内角和来解决这个问题。首先，我们计算三角形的内角和： $\angle A + \angle B + \angle C = 180^\circ$ 。 $50^\circ + 20^\circ + \angle ADC = 180^\circ$ 。接下来，我们需要找到另一个角度，即 $\angle ADC$ 。我们可以通过将内角和减去已知角度来得到它： $\angle ADC = 180^\circ - \angle A - \angle C$ 。 $\angle ADC = 180^\circ - (50^\circ + 20^\circ)$ 。 $\angle ADC = 180^\circ - 70^\circ$ ， $\angle ADC = 110^\circ$ 。所以，三角形 $\triangle ABE$ 与 $\triangle ACD$ 的另一个角度是 $\angle ADC = 110^\circ$ 。

Judgment Result:

Role: Assistant

The relevant paragraph in the Model Response that provides the answer is

所以，三角形 $\triangle ABE$ 与 $\triangle ACD$ 的另一个角度是 $\angle ADC = 110^\circ$

Understand the question in Chinese. The Chinese model response means: Therefore, another angle of triangle $\triangle ABE$ and $\triangle ACD$ is $\angle ADC = 110^\circ$.

Next, let's analyze whether the correct answer has been given. The reference answer 110° is a numerical answer, and the output given in the model response is 110° . The consistency of the answers is: <Yes>.

Figure 14: A instruction example to finetune GradeGPT.

puts and translating them into English. Compared to having the language model directly predict the <Yes>/<No> judgment labels, directly using Chinese Judge-of-Chain to construct Judge-of-Chain improves the model's performance in answer evaluation. However, they didn't yield good results. The performance disparity may stem from an imbalance in the quality of Chinese and English components within some bilingual base models due to the training corpus.

Our findings show that fine-tuning with Cross-Lingual-Judge-of-Chain for detailed thought chain refinement significantly improves the performance of open-source models in outcome analysis tasks. Additionally, we discovered that using bilingual thought chains instead of Chinese-only thought chains for base model fine-tuning effectively enhances performance in outcome determination tasks. By using and synthesizing the instructions in the form of Cross-Lingual-Judge-of-Chain that we designed, we are able to efficiently distill the answer reviewing capabilities of GPT-4.

Ablation On Instruction Data Source The instruction data for Cross-Lingual Judge-of-Chain Prompts comes from outputs of various LLMs on CMMaTH. We conducted ablation experiments on the sources of instruction data in Table 10, which showed the impact of using different LLM models in constructing diverse and effective instruction

LLM	$Acc_{outcome}$
Qwen-7B-Chat(4-Shot)	35.1
+Naive Outcome Finetune	51.5
+Judge-of-Chain	65.3
+Cross-Lingual-Judge-of-Chain	85.1
Qwen-14B-Chat(4-Shot)	43.7
GradeGPT(14B)	96.1
GPT-4(4-Shot)	97.2

Table 9: Ablation study on the instruction fine-tuning of GradeGPT commands

Model Response Source	$Acc_{outcome}$
LLaVA-v1.5 response	77.2
+InterLM-XComposer2-VL response	83.1
+CogAgent-Chat response	84.2
+Yi-VL-34B response	85.1

Table 10: Ablation study on the instruction fine-tuning of GradeGPT commands. All experiments were conducted using the GradeGPT (7B) version.

data.

$$Acc_{outcome} = \frac{I(GradeGPT(R_i), Overcome_{GT})}{N_{response}} \times 100 \quad (2)$$

H Prompt Details

H.1 Prompt For Step Response Generation

When evaluating hallucinations during the assessment process, we use a few-shot prompt format

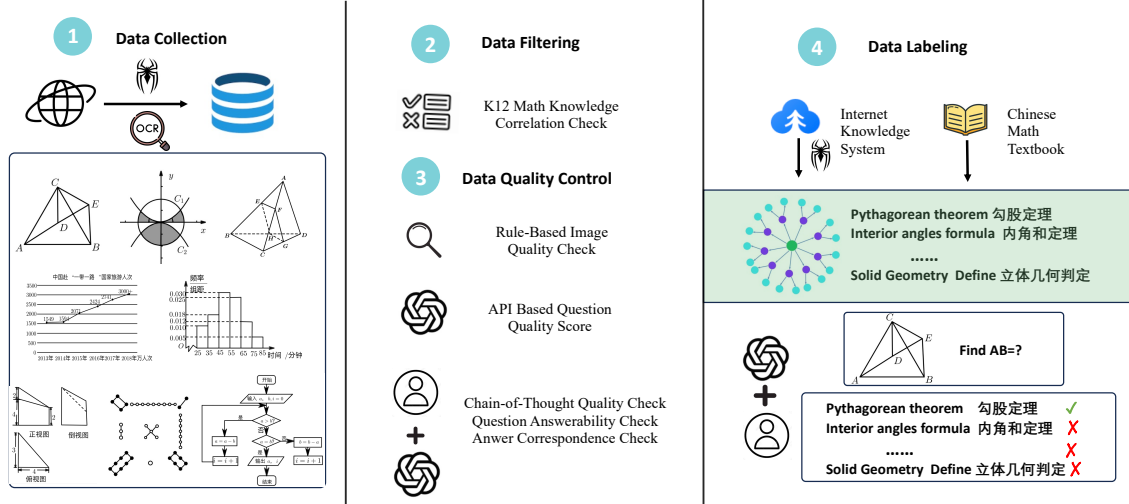


Figure 15: Overall Data Collection Pipeline of CMMaTH.

Element	Prompt
system-prompt	你是一个数学题解题助手，你的输入是一道数学题，以及这道题目的图像，你的任务是输出这道题目的求解思路以及答案。输出格式是一步一步的方式。
Example1	Question: 小明有2000元，一个苹果2元，小明买了10个苹果。小明还有多少钱。 Model response: 首先计算小明买苹果的开销为2乘10等于20元。<STEP1> 用2000元减去所有的开销，2000-20=1980。<STEP2> 所以最终的答案为1980元。<STEP3>
Example2	Question: 如图，在$\triangle ABC$中，$\angle C=90^\circ$，$\angle B=30^\circ$，以A为圆心，任意长为半径画弧分别交AB、AC于点M和N，再分别以M、N为圆心，大于$MN/2$的长为半径画弧，两弧交于点P，连结AP并延长交BC于点D，则下列说法中正确的个数是（ ）①AD是$\angle BAC$的平分线；②$\angle ADC=60^\circ$；③点D在AB的中垂线上；④$BD=2CD$。A.4 B.3 C.2 D.1 Model response: 首先确定$\triangle ABC$的几何性质。<STEP1>$\angle C=90^\circ$，$\angle B=30^\circ$，$\angle A=120^\circ$。<STEP2>$\triangle ABC$是一个等腰三角形。<STEP3>根据等腰三角形的性质，AD是$\angle BAC$的平分线。<STEP4>①AD是$\angle BAC$的平分线，正确。<STEP5>②$\angle ADC=60^\circ$，错误。<STEP6>③点D在AB的中垂线上，错误。<STEP7>④$BD=2CD$，正确。<STEP8>因此，正确的个数为4。<STEP9>答案为A.4。<STEP10>请你根据这个例子，解决下面的数学题。问题：在$\triangle ABC$中，$\angle C=90^\circ$，$\angle B=30^\circ$，以A为圆心，任意长为半径画弧分别交AB、AC于点M和N，再分别以M、N为圆心，大于$MN/2$的长为半径画弧，两弧交于点P，连结AP并延长交BC于点D，则下列说法中正确的个数是（ ）①AD是$\angle BAC$的平分线；②$\angle ADC=60^\circ$；③点D在AB的中垂线上；④$BD=2CD$。A.4 B.3 C.2 D.1 求解步骤：首先确定$\triangle ABC$的几何性质。$\angle C=90^\circ$，$\angle B=30^\circ$，$\angle A=120^\circ$。$\triangle ABC$是一个等腰三角形。根据等腰三角形的性质，AD是$\angle BAC$的平分线。①AD是$\angle BAC$的平分线，正确。②$\angle ADC=60^\circ$，错误。③点D在AB的中垂线上，错误。④$BD=2CD$，正确。因此，正确的个数为4。答案为A.4。

Table 11: Prompt for all model to generate step-by-step answer.

to elicit step-by-step outputs from the model as showed in Table 11.

H.2 Prompt For GradeGPT

We also listed the prompts used by GradeGPT in Table 12.

H.3 Prompt For Cross-Lingual Prompting and En-CoT

We have listed the specific prompts used for En-CoT in Table 13 and Cross-Lingual Prompt during actual execution in Table 14. Unlike the original

Cross-Lingual Prompt paper, for experimental simplicity, we only adopted a single-turn format. However, this suffices to illustrate the varying inferential capabilities across different languages in current LMMs.

Element	Prompt
system-prompt	You are an answer evaluation assistant. Based on the following questions and reference answers, determine if the model's output is the same answer. When making the determination, please provide the analysis process in the following format. Note, for multiple-choice questions, it is only correct if the output includes all the corresponding options. (For example: If the reference answer is: A, C, and the model response is: C, the answer consistency is: No) Note, if the model response does not specify a particular option, and its correctness cannot be determined, the answer consistency should be: No.
Example	Question: 如图所示, AD是$\triangle ABC$的中线, 点E在AC上, BE交AD于点F. 当$\frac{AF}{FD} = \frac{1}{2}$时, $\frac{AE}{AC} =$ Model Response: 首先, 我们需要找到中线AD, 然后找到点E在AC上的位置。找到中线AD, 可以通过勾股定理求解, $AD^2 = AD \times AC^2$。设$AD = x, AC = y$, 则有$x^2 = xy^2$。<STEP1> 解这个方程, 我们得到 $y = \sqrt{x}$。现在我们知道点E在AC上, 所以E的横坐标为y, 纵坐标为x。 接下来, 我们需要找到点F在BD上的位置。由于F点在BD上, 我们可以设F点在BD的一半, 即$F = (x + y)/2$。<STEP2> 现在我们可以计算AF和FD的长度。 $AF = x - F, FD = FD - F$ 将已知的x和y代入公式, 得到: $AF = x - (x + y)/2, FD = (x + y)/2 - (x + y)/2$ 简化得到: $AF = x - x, FD = x - x$ 这意味着$AF = FD$。<STEP3> 现在我们可以求解题目中的条件: $AE/AC = 1/2$ 将已知的AF和FD代入公式, 得到: $AE/AC = (x - x)/(\sqrt{xy})$化简得到: $AE/AC = 1/2$所以, 答案为$AE/AC = 1/2$ <STEP4> Reference Response: $\frac{4\pi}{3} - 2\sqrt{3}$ Judge Result: The relevant paragraph in the Model Response that provides the answer is 所以, 答案为$AE/AC = 1/2$。 Understand the question in Chinese. The Chinese model response means: Therefore, the answer is $AE/AC = 1/2$. Next, let's analyze whether the correct answer has been given. The reference answer $\frac{1}{5}$ is a numerical answer, and the output given in the model response is $1/2$. The consistency of the answers is: <No>.

Table 12: Detail prompt for GradeGPT and it's instruction construction prompt by GPT4.

Element	Prompt
system-prompt	You are a math problem-solving assistant. Your input is a math problem and an image of the problem. Your task is to provide the solution steps and the final answer. The output format should be step-by-step.
Example	Question: Xiaoming has 2000 yuan, an apple costs 2 yuan, Xiaoming bought 10 apples. How much money does Xiaoming have left? Model response: First, calculate the cost of the apples Xiaoming bought, which is 2 times 10 equals 20 yuan. <STEP1> Subtract the total cost from 2000 yuan, $2000 - 20 = 1980$. <STEP2> So the final answer is 1980 yuan. <STEP3>

Table 13: Prompt for En-CoT in Our Cross-Lingual Experiment.

Element	Prompt
system-prompt	Please act as an expert in multi-lingual understanding in Chinese. After understanding, you should act as an expert in arithmetic reasoning in English. Let's resolve the task you understand above step-by-step! Finally, you should format your answer as 'Answer: [num]'.
Example	Request: 玛丽比琼小两岁, 琼比杰莎大五岁。如果杰莎20岁, 这三个女孩的年龄总和是多少? Reason Result: Let's understand the task in English step-by-step! 1. "玛丽比琼小两岁" means "Mary is two years younger than Joan." 2. "琼比杰莎大五岁" means "Joan is five years older than Jessie." 3. "如果杰莎 20 岁" means "If Jessie is 20 years old." Then, Let's solve the task step-by-step: 1. According to the first sentence, Mary is two years younger than Joan. Let's suppose Joan's age is J, then Mary's age is J - 2. 2. According to the second sentence, Joan is five years older than Jessie. If Jessie's age is x, then Joan's age is x + 5. Answer: 68.

Table 14: Detail prompt for Cross-Lingual Prompt(CLP) Reason technology for our Experiment.

Model Name	Generation Parameters	Comments
MAmmoTH-70B	temperature=0.2, max_new_tokens=2048	vllm package
InternLM-Math	temperature=0.2, max_new_tokens=2048	vllm package
MetaMath-70B	temperature=0.2, max_new_tokens=2048	vllm package
DeepSeek-Math-7B	temperature=0.2, max_new_tokens=2048	vllm package
Llama-2-70B	do_sample=True, top_k=0.5, top_p=0.5, max_tokens=512	vllm package
Baichuan-13B	temperature=0.2, max_new_tokens=2048	vllm package
Qwen-14B	temperature=0.2, max_new_tokens=2048	vllm package
llava-7B-V1.5	temperature=0.2, max_new_tokens=2048	llava package
Yi-VL-34B	temperature=0.2, max_new_tokens=2048	Huggingface
LLaVA-NEXT-34B	temperature=0.2, max_new_tokens=2048	Huggingface
LLaVA-OneVision	temperature=0.2, max_new_tokens=2048	Huggingface
CogAgent-Chat	temperature=0.2, max_new_tokens=2048	Huggingface
Qwen2-VL	temperature=0.2, max_new_tokens=2048	Huggingface
InternVL2	temperature=0.2, max_new_tokens=2048	Huggingface
InterLM-XComposer2-VL	temperature=0.2, max_new_tokens=2048	Huggingface
CogVLM	temperature=0.2, max_new_tokens=2048	Huggingface
GPT-4	temperature=0.2, max_tokens=2048	version="gpt-4-1106-preview"
Gemini	temperature=0.2, max_tokens=2048	version="gemini-1.5-Pro-2023-05-15"
Claude	temperature=0.2, max_tokens=2048	version="claude-3.5-sonnet-2024-05-24"
GPT-4V	temperature=0.2, max_tokens=2048	version="gpt-4-vision-2023-05-15"
GPT-4o	temperature=0.2, max_tokens=2048	version="gpt-4o-2024-05-14"

Table 15: The hyperparameters for the models used in the evaluation are detailed. When the "comments" section includes the format *model = ""*, it signifies that the model was loaded from the transformer package. The vLLM package indicates that models are implemented by the vLLM package, where more details can be found in <https://github.com/vllm-project/vllm>. For models other than OpenAI’s GPT, custom codes were utilized for evaluation unless specified otherwise in the comments.

Model	Name
Llama-2-70B	meta-llama/Llama-2-70b-hf
Baichuan-13B	baichuan-inc/Baichuan2-13B-Chat
Qwen-14B	Qwen/Qwen-14B-Chat
InternLM-Math	internlm/internlm2-math-20b
DeepSeek-Math-7B	deepseek-ai/deepseek-math-7b-instruct
MetaMath-70B	meta-math/MetaMath-70B-V1.0
MAMmoTH2-8x7B	TIGER-Lab/MAMmoTH2-8x7B
Yi-VL-34B	01-ai/Yi-VL-34B
Qwen2-VL	Qwen/Qwen2-VL-7B-Instruct
LLaVA-v1.5	liuhaotian/llava-v1.5-13b
LLaVA-NEXT-8B	llava-hf/llama3-llava-next-8b-hf
LLaVA-OneVision	lmms-lab/llava-onevision-qwen2-7b-ov
InternVL2-8B	OpenGVLab/InternVL2-8B
InterLM-XComposer2-VL	internlm/internlm-7b
CogVLM	THUDM/cogvlm-chat-hf
CogAgent	THUDM/cogagent-vqa-hf
Gemini	<Gemini 1.5 Pro>
Claude	<Claude 3.5 Sonnet>
GPT4o	<GPT4o-20240201>

Table 16: LLMs used in our experiments and their corresponding names in Huggingface Hub.

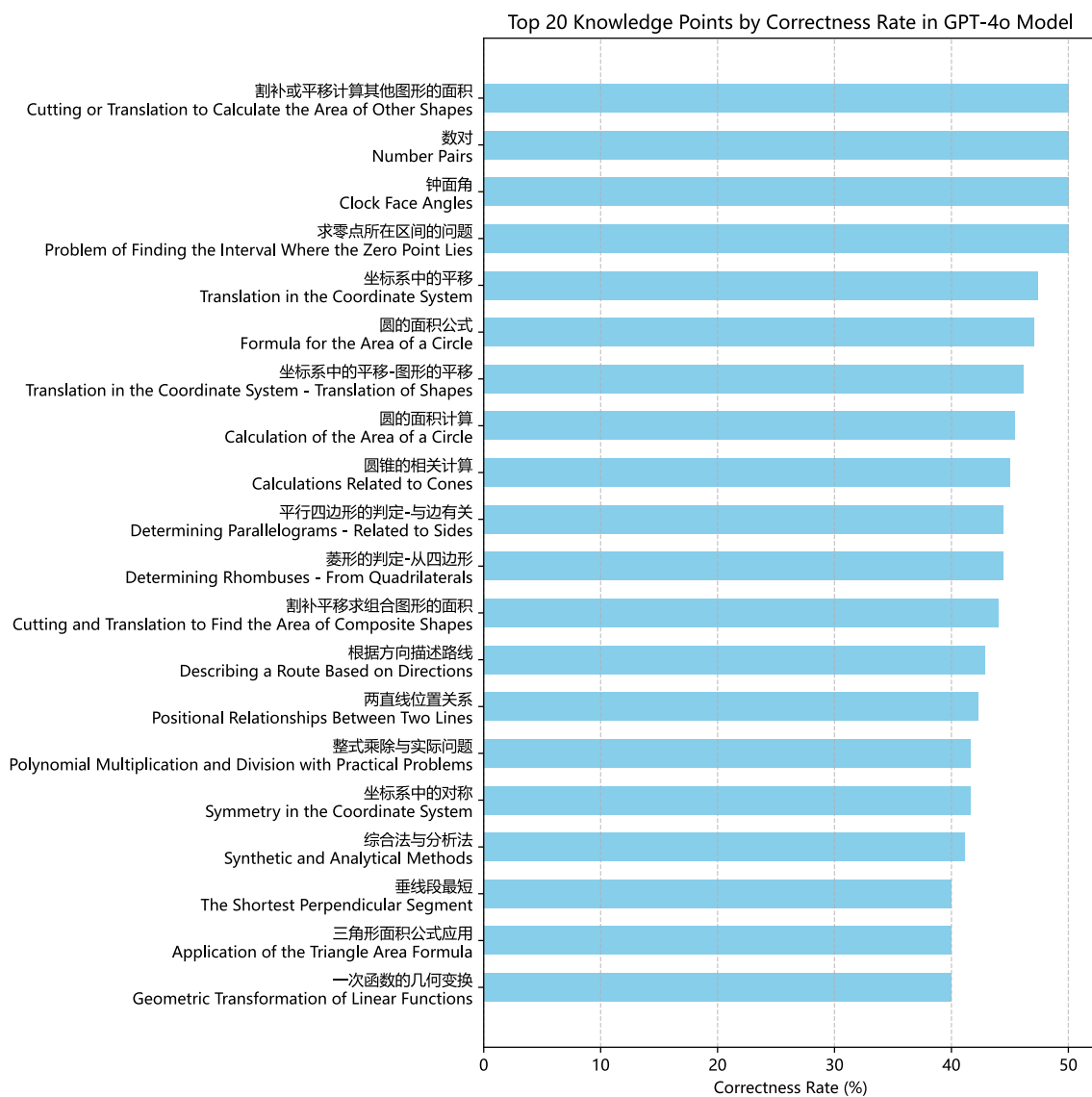


Figure 16: Top 20 Knowledge Point Correctness of GPT-4o.

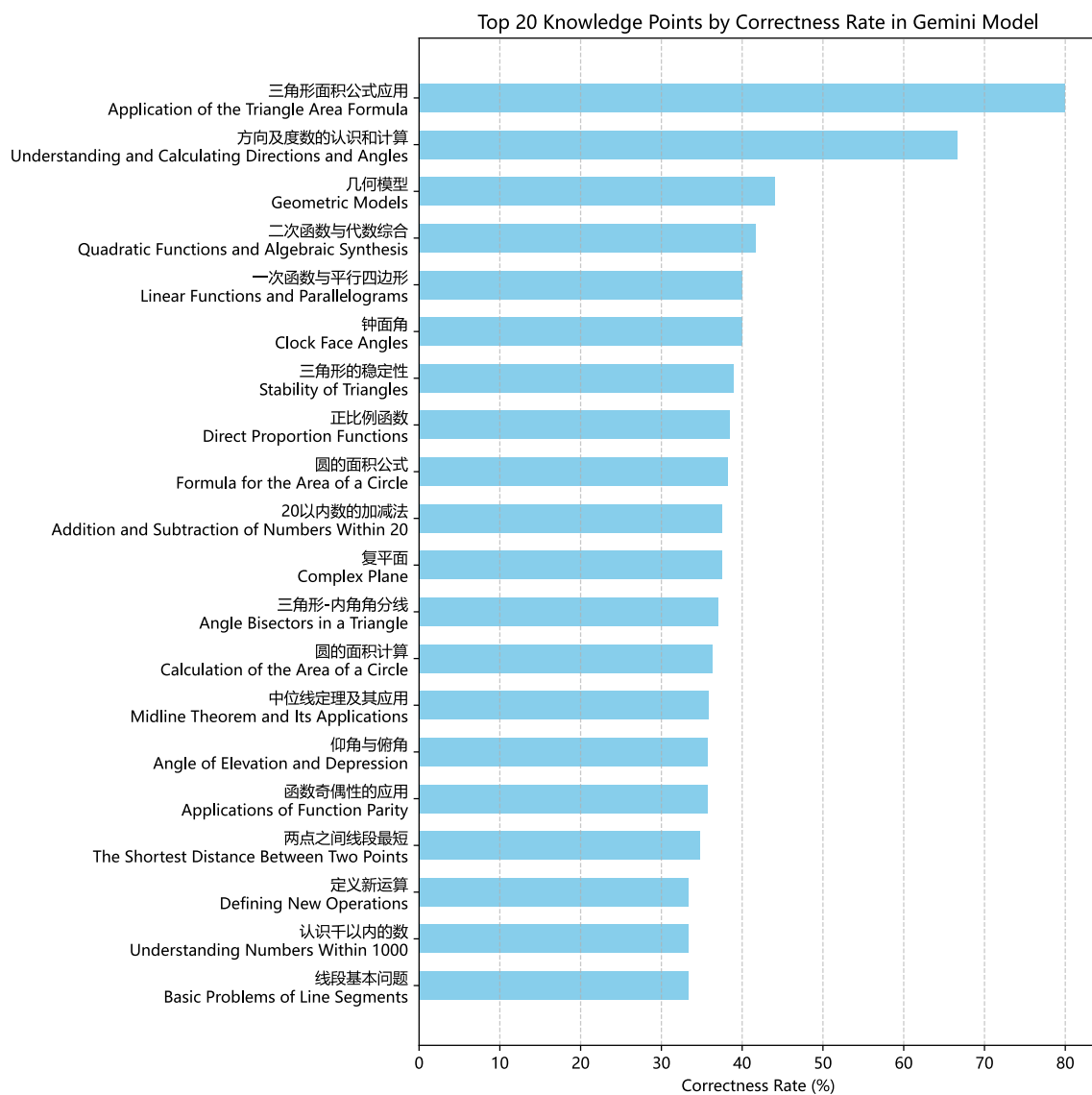


Figure 17: Top 20 Knowledge Point Correctness of Gemini-Pro.

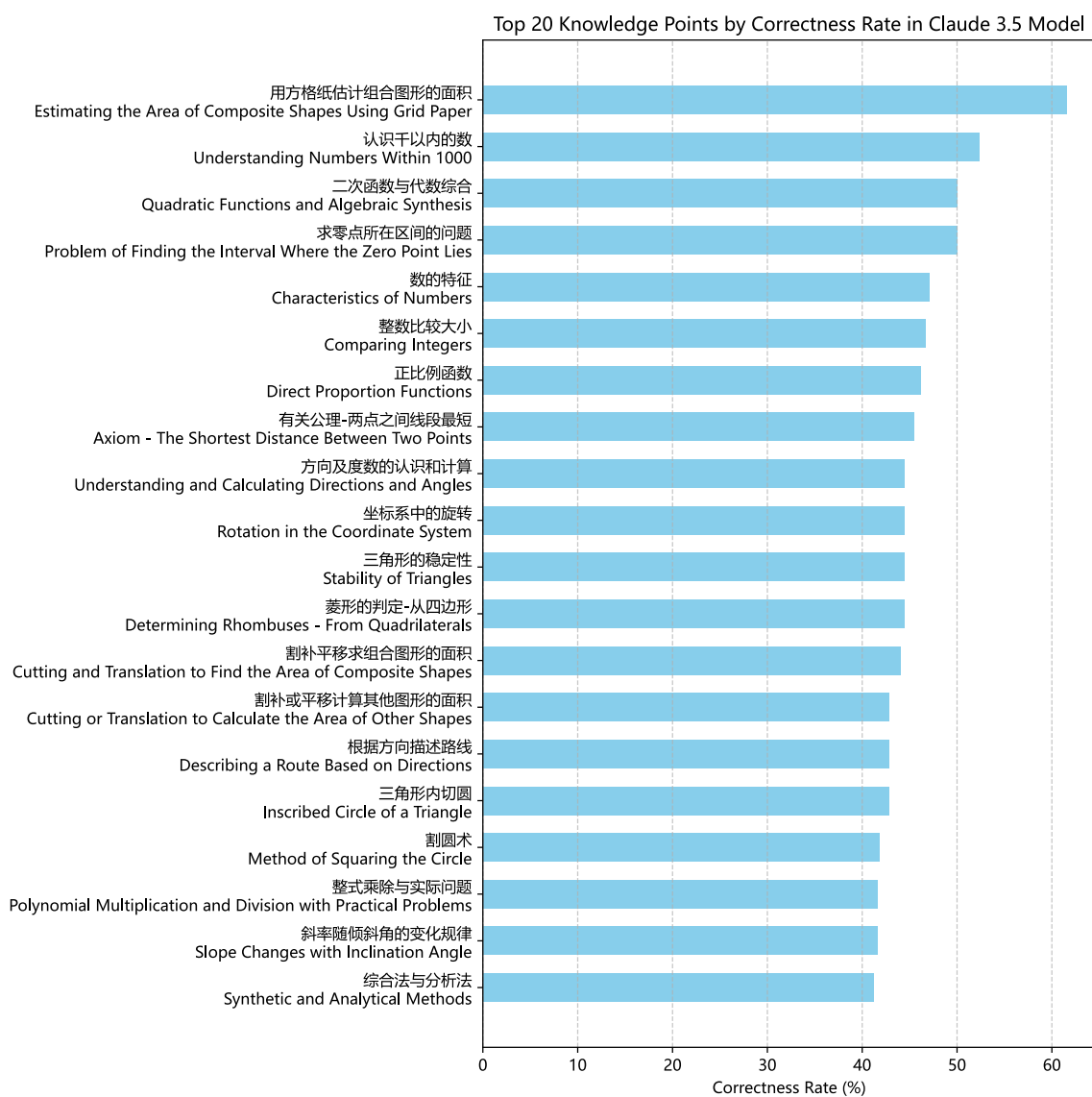


Figure 18: Top 20 Knowledge Point Correctness of Claude-3.5.

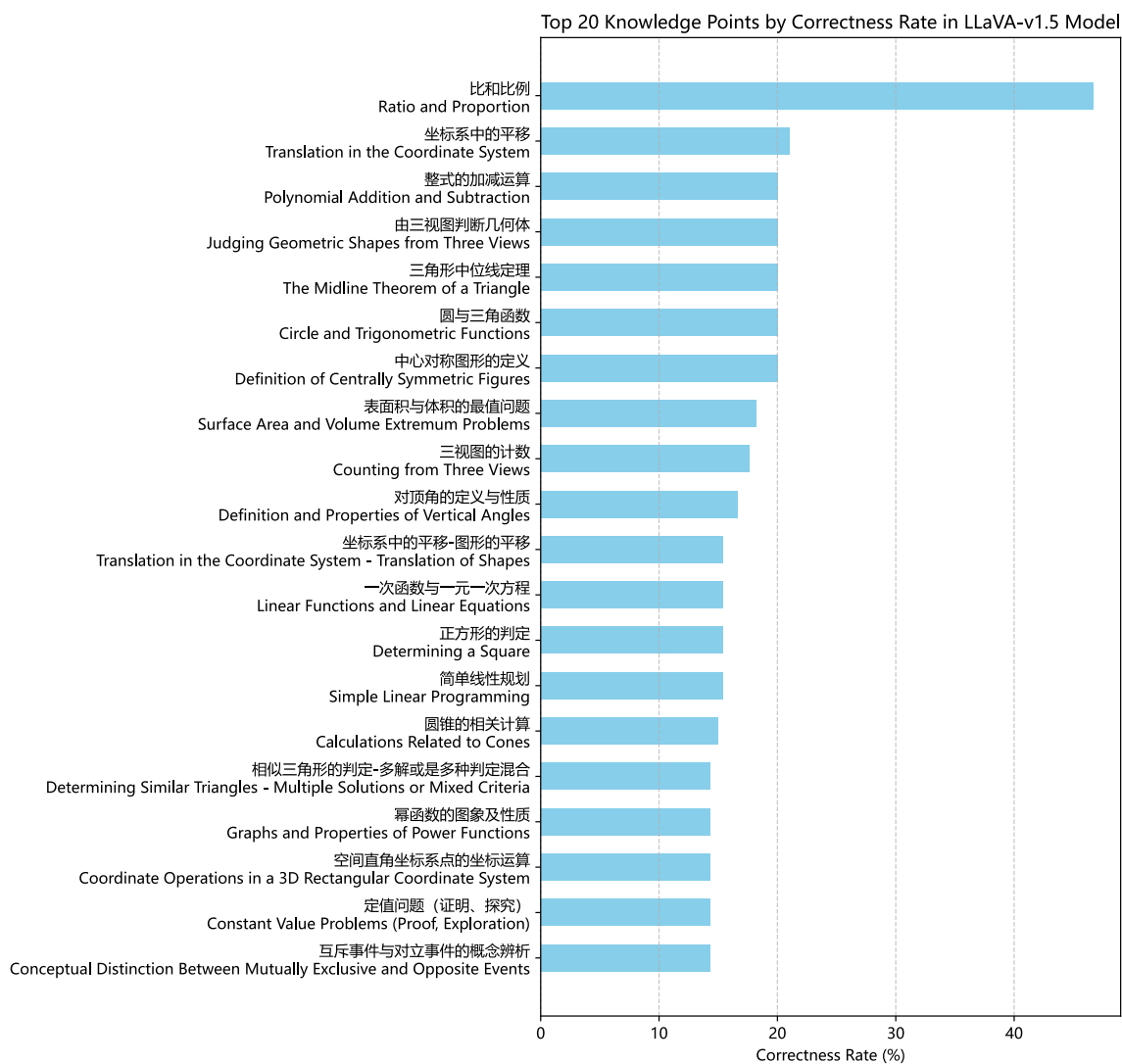


Figure 19: Top 20 Knowledge Point Correctness of LLaVA-v1.5.



Figure 20: Poor Solving Performance Knowledge Point Cloud of Gemini-Pro, Claude-3.5, GPT-4o and LLaVA-v1.5.

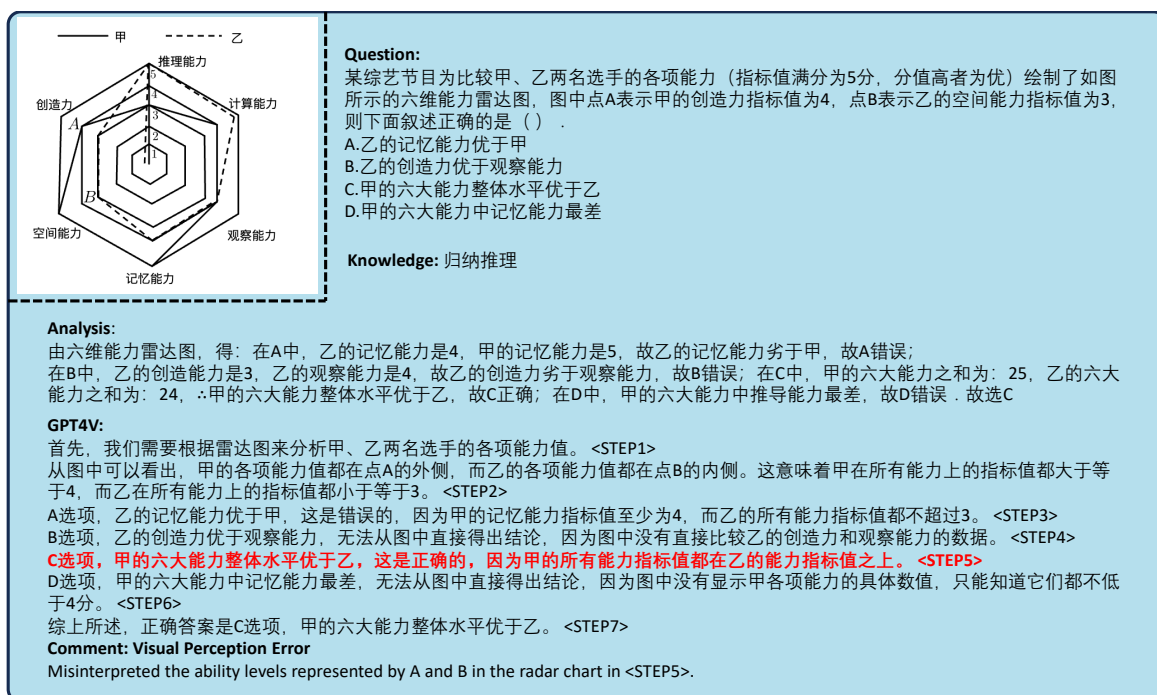


Figure 21: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

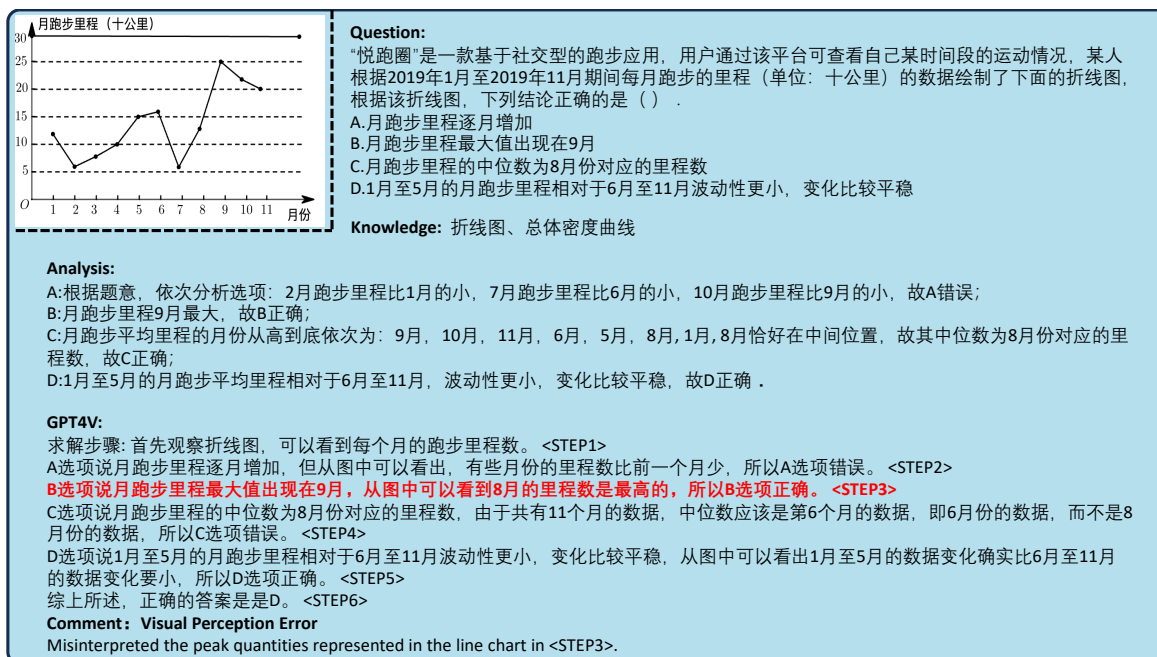


Figure 22: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

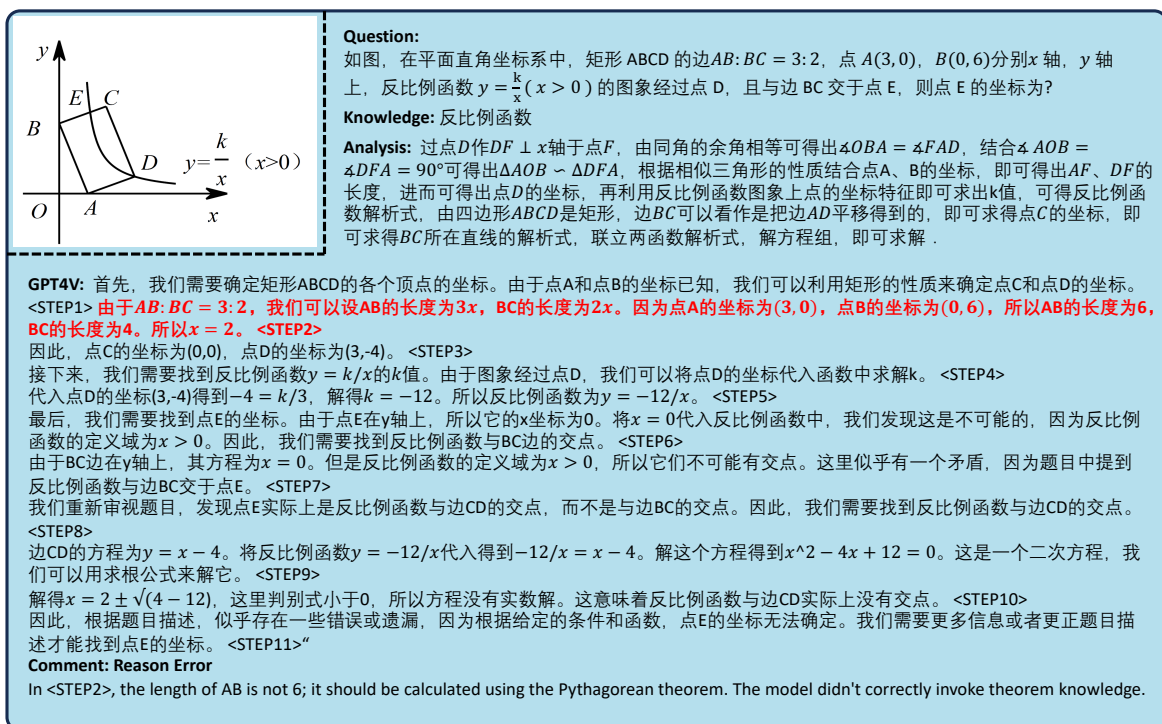


Figure 23: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

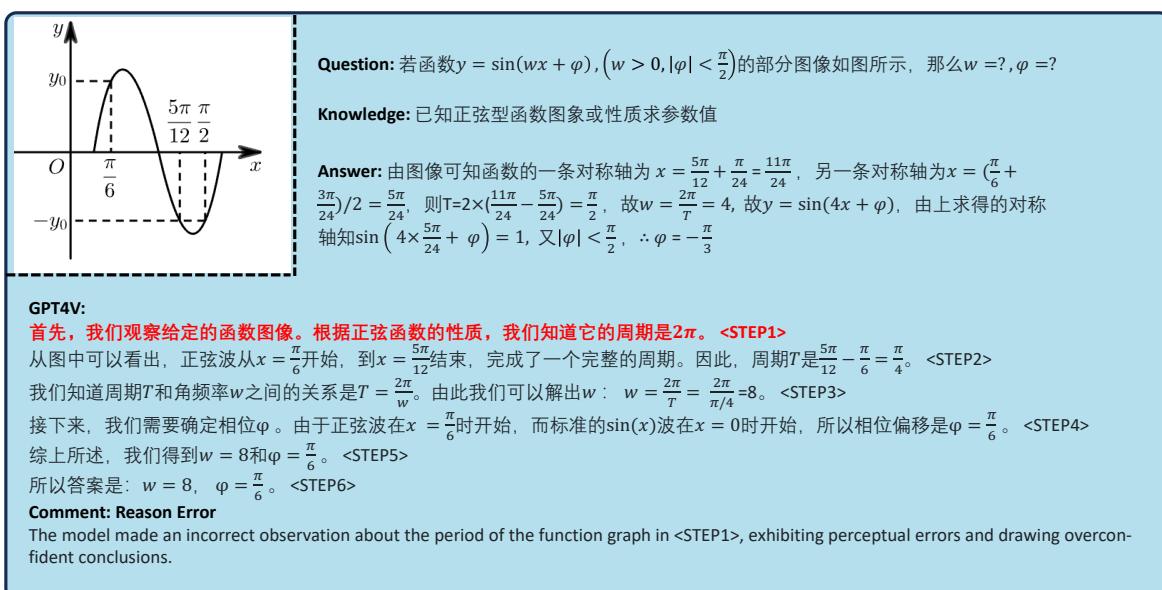


Figure 24: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

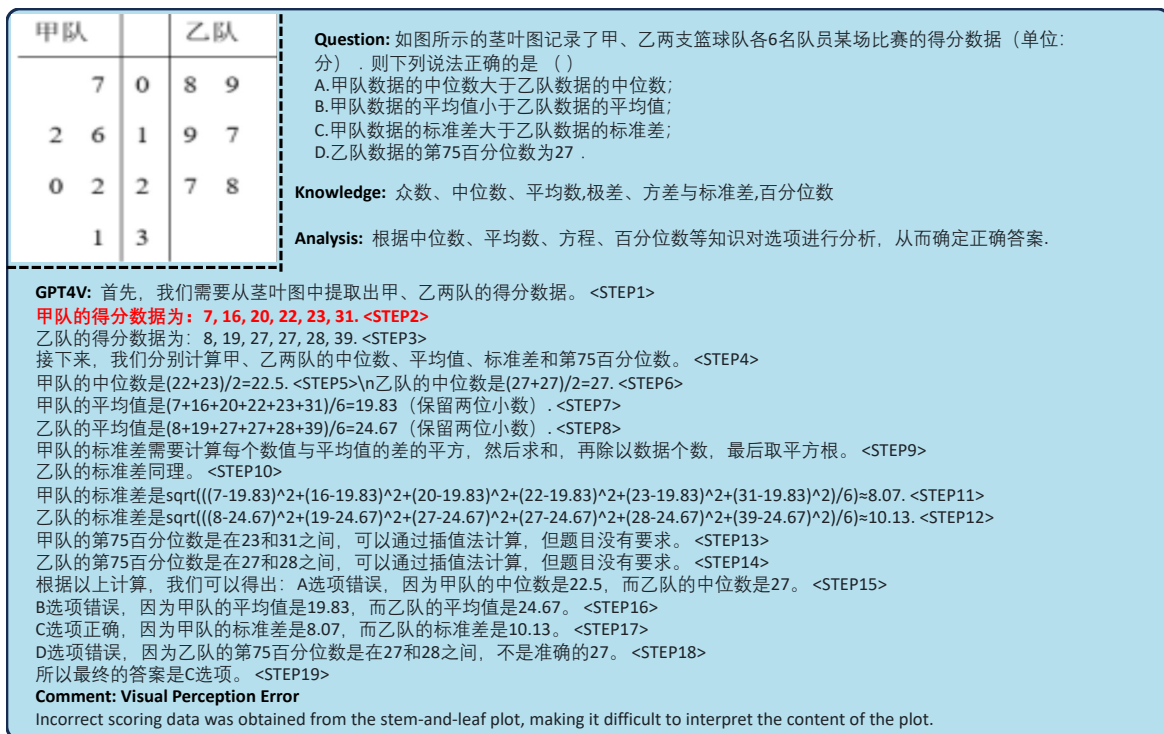


Figure 25: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

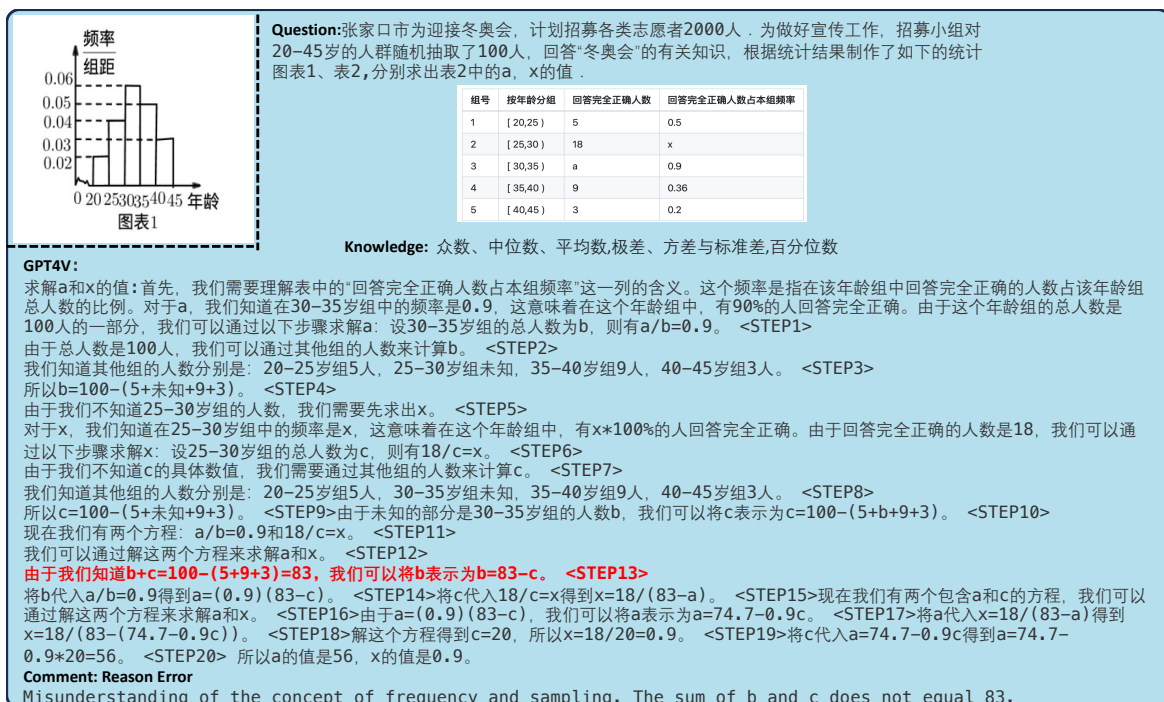


Figure 26: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

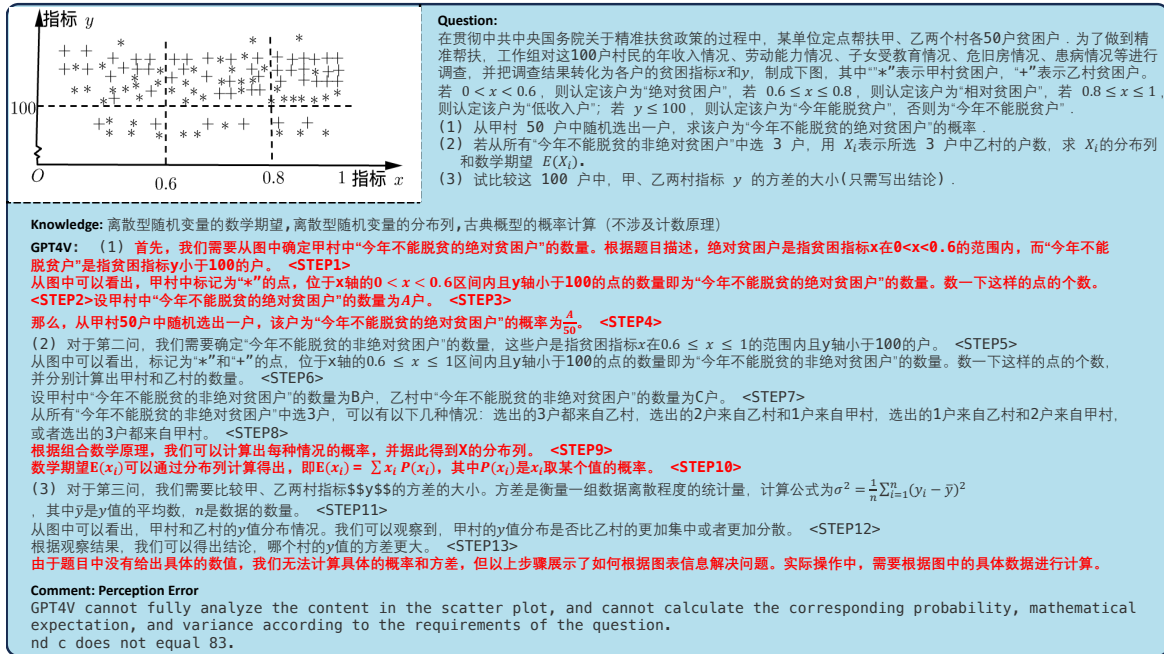


Figure 27: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

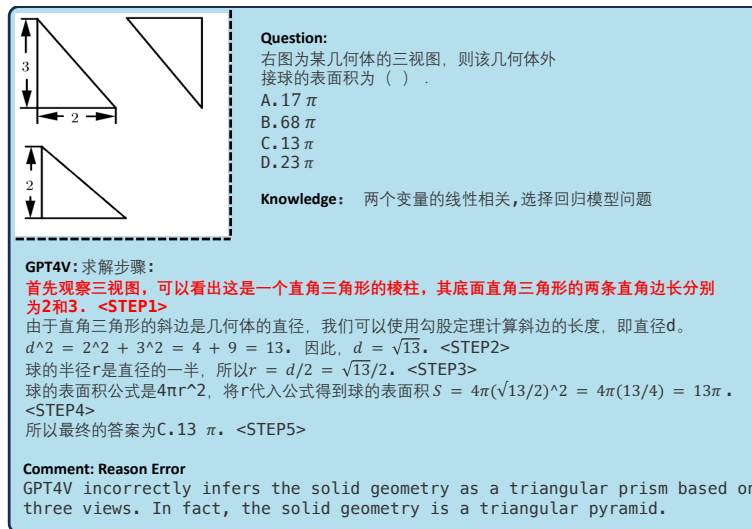


Figure 28: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

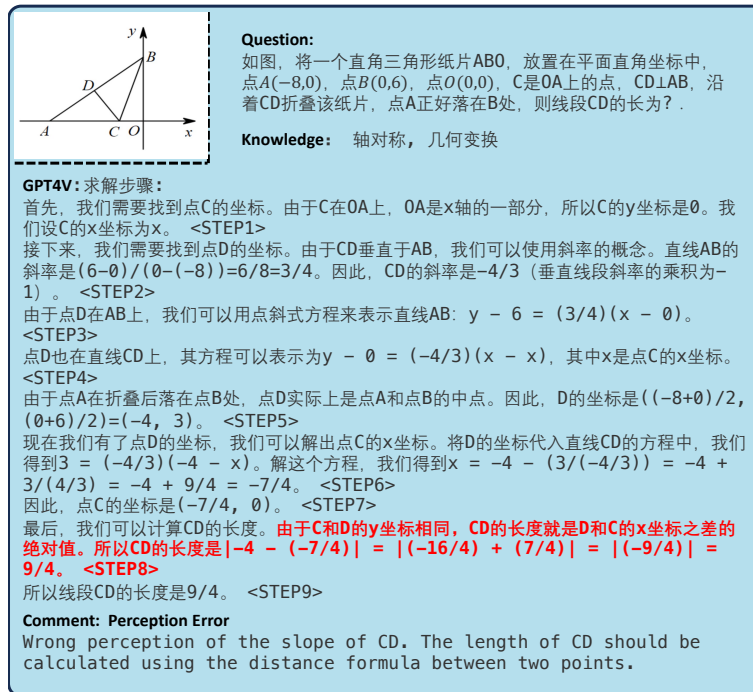


Figure 29: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.

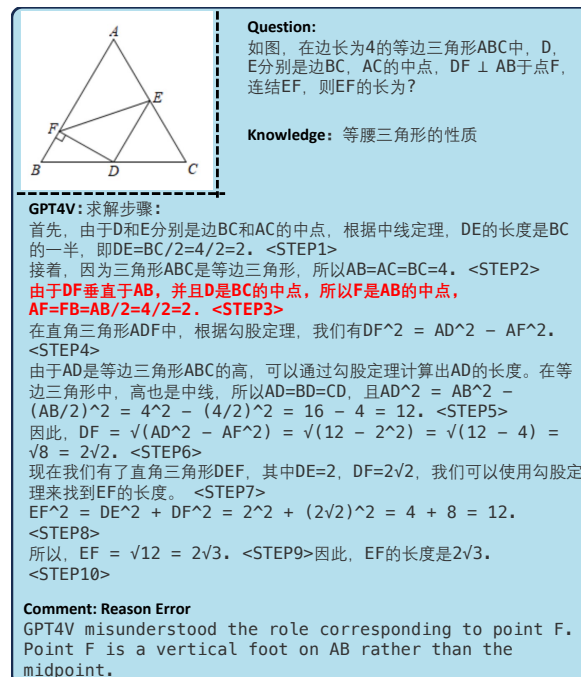
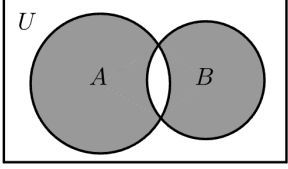


Figure 30: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.



Question:已知全集 $U = R$ ，集合 $A = \{x|0 \leq x \leq 2\}$ ， $B = \{x|x^2 - x > 0\}$ ，则图中的阴影部分表示的集合为？

A. $\{x|x \leq 1 \text{ 或 } x > 2\}$
 B. $\{x|x < 0 \text{ 或 } 1 < x < 2\}$
 C. $\{x|1 \leq x < 2\}$
 D. $\{x|1 < x \leq 2\}$

Knowledge: 集合的概念与表示方法

GPT4V: 求解步骤:
 首先解集合B的不等式 $x^2 - x > 0$ ，可以分解为 $x(x - 1) > 0$ ，得到 $x > 1$ 或 $x < 0$ 。 <STEP1>
 集合A是从0到2的闭区间，即 $0 \leq x \leq 2$ 。 <STEP2>
图中阴影部分表示的是集合A和B的交集，即 $A \cap B$ 。 <STEP3>
 根据集合A和B的定义，交集部分是 $1 < x \leq 2$ 。 <STEP4>
 所以最终的答案是D. $\{x|1 < x \leq 2\}$ 。 <STEP5>

Comment: Reason Error
 Wrong understanding of the contents of the shaded part. The intersection of sets A and B does not represent the area of the shaded part.

Figure 31: Case from GPT-4V/o. The red ones are marked as generated inference hallucinations.